Design-Based Justification for Clustering with Multiway Assignment

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Abstract

In causal inference, it is often not obvious what standard errors are researcher should report, and what dimensions a researcher should cluster the standard errors by, if at all. This paper provides a design-based perspective to when researchers should cluster when doing inference on the sample average treatment effect (ATE). By using the potential outcomes model, the researcher can be agnostic about the data generating process for the outcome and only has to consider randomness in the assignment mechanism to treatment, which is often better understood. The practical takeaway is that, when we are interested in the sample ATE, the only consideration on whether to cluster and which dimensions to cluster on is determined by assignment. Notably, this does not depend on the structure of the error term as is commonly argued in literature. Cluster on a particular dimension if and only if it affects assignment. Standard errors are conservative if we cluster on more dimensions than we have to. I illustrate the findings with a simple simulation and show consistency and asymptotic normality of standard estimators in this design-based framework.

1 Introduction

When doing causal inference of the effect of treatment W on outcome Y, it is often not obvious what standard errors researchers should report. A common option is to use the heteroskedasticity-robust standard errors attributed to Eicker (1967), Huber (1967) and White (1980) (henceforth EHW). When there are dependence structures in the data that casts doubts on the independence over

observations, the cluster-robust standard errors attributed to Liang and Zeger (1986) (henceforth LZ) is often employed. By clustering standard errors, we allow arbitrarily correlated standard errors for units sharing a cluster. However, it is not obvious what dimensions the researcher should cluster on. For concreteness, consider the example in Dube et al. (2010) who are interested in the effect of minimum wages on employment. They have data over many states and border segments, and restaurants are their units of observation. It is unlikely that restaurants in the same border segment are iid, so it might be sensible to cluster by border segment due to common labor market shocks. However, if we are worried about labor market shocks, why do we not cluster by restaurant chain, for instance? Alternative arguments could also suggest some correlation based on the age of the manager, or the county that the restaurant operates in. In light of the difficulty in choosing which dimensions to cluster on, this paper aims to shed light on this complication.

Multiway clustering occurs when researchers clusters on more than one dimension, and errors are allowed to be arbitrarily correlated whenver observations share at least one cluster. There are many examples of multiway clustering in empirical work. For instance, Hersch (1998) is interested in the relationship between injury risk and wages, and the author clusters by industry and occupation. Rose and Engel (2002) have a bilateral trade model that has observations in country pairs, so one might want to have two-way clustering for the countries involved in each pair. Gruber and Madrian (1995) are interested in relationship between health insurance availability and retirement. Clusters involved are state-year cell and household. This is an issue in almost any matched data e.g., employer-employee matched data. However, it is hard to justify why we want to cluster over some dimensions but not others. Current justification in literature is model-based, while there are no design-based asymptotic arguments on how multiway clustering captures the true extent of uncertainty to the best of my knowledge, which is a novelty of this paper.

To clarify the difference between model-based and design-based arguments, consider a binary treatment $W_i \in \{0, 1\}$, and outcome given by $Y_i = \alpha + \tau W_i + \varepsilon_i$. Model-based uncertainty stems from the concern in Moulton (1986) that the errors may be correlated within clusters, so we would think of outcomes Y_i as random while W_i are exogenous objects. In contrast, design-based uncertainty uses a potential outcome model so potential outcomes $Y_i(w)$ indexed by treatment status w are nonstochastic objects that we condition on, while W_i are stochastic objects, so randomness occurs in assignment to treatment. Consequently, in the design-based framework, randomness in ε_i only comes from assignment uncertainty. These error terms are functions of potential out-

comes, with $\varepsilon_i(w) = Y_i(w) - \frac{1}{N}\sum_{i=1}^{N} Y_i(w)$ where N is the size of the population of interest, so $\varepsilon_i = W_i\varepsilon_i(1) + (1 - W_i)\varepsilon_i(0)$. By using the potential outcomes framework, this paper speaks to the growing literature on heterogeneous treatment effects.

In causal inference, it is often more sensible to consider randomness in assignment instead of randomness in the outcome. The objective of causal inference is to learn about causal effect $Y_i(1) - Y_i(0)$, so it seems unusual that we want to make assumptions about the outcome in order to learn about the outcome. Instead, it is more sensible to make assumptions about the assignment process that drives the observed outcome. This should lead us to prefer design-based inference to model-based inference. Further, in many empirical applications, including natural experiments and randomized controlled trials, researchers understand the assignment mechanism much better than the sampling mechanism or the underlying data-generating process for the outcome. By focusing on the sample average treatment effect (ATE), defined as $\tau := \frac{1}{N}\sum_{i=1}^{N}(Y_i(1) - Y_i(0))$, this approach exploits the information that the researchers already know and are confident of, while remaining agnostic about aspects of the empirical design that are less understood. Focusing on the sample ATE also allows us to abstract from uncertainty arising from how observations are sampled from some superpopulation.

There are several reasons why we would be interested in the sample ATE. (1) With population data (e.g., administrative data or census data) available to researchers, this is equivalent to doing inference on the ATE for the population with population data - a situation that is becoming increasingly relevant. (2) Sample ATE is a minimal benchmark for identification and inference, before making statements about external validity. (3) There is no need to worry about sampling or sample selection, because the sample we observe is the set of observations that we are already interested in. (4) This takes internal validity seriously. There is often much critique about the external validity of causal statements, so researchers often aim to make a statement on the internal validity of the conclusions first. If we do take internal validity seriously, we should report standard errors designed for internally valid conclusions, rather than interpret the estimates as objects that are internally valid while reporting standard errors that are justified for an externally valid object.

Multiway clustering in a design-based framework can be justified through multiway assignment. Two-way assignment is determined by $W_i = f(A_{G_i}, B_{H_i}, v_i)$, where f is some generic function that we can be agnostic about, G_i, H_i denotes clusters on the G and H dimension that observation i

belongs to, and v_i is an individual-specific error. Here, v_i is drawn iid over observations, and A_g, B_h are drawn iid over clusters on the G and H dimensions respectively. In this generic framework, observations of W_i are thus arbitrarily correlated if they share at least one cluster except on edges cases (e.g., $var(A_g) = 0$), and are independent of each other if they have no clusters in common. This mimics the model-based framework for multiway clustering.

This paper makes two major contributions. The first contribution is practical. I show how multiway clustering captures the true extent of uncertainty when there is multiway assignment in a design-based framework. The practical takeaway here is that, when we are interested in the sample ATE, the only consideration on whether to cluster and which dimensions to cluster on is determined by assignment. Cluster on a particular dimension if and only if it affects assignment. Standard errors are conservative if we cluster on more dimensions than we have to. The second contribution is theoretical. I prove consistency and asymptotic normality of standard estimators in a design-based framework. These limit theorems are proven without separate exchangeability, which is a condition often assumed in standard limit theorems with multiway clustering, since separate exchangeability fails once we move from a model-based framework to a design-based framework. Existing literature on multiway clustering also relies on separate exchangeability of cluster sizes, which is not required in this paper. Instead, this paper imposes much weaker conditions on cluster sizes, which accommodates situations where the convergence rate is nonstandard (i.e., slower than $N^{-1/2}$), building on the recent asymptotic theory for oneway clustering.

1.1 Literature Review and Contribution

This paper can be viewed as the first that bridges that gap between design-based inference and multiway clustering. This section highlights how this paper contributes to the design-based literature, and how it addresses challenges unique to multiway clustering. There is a recent growing literature on design-based inference, which includes Abadie et al. (2017), Abadie et al. (2020), Rambachan and Roth (2020), and Xu (2021). The econometric literature on multiway clustering proceeds from a model-based framework. Cameron et al. (2011) (henceforth CGM) propose a closed-form variance formula for multiway clustering by extending the LZ estimator. They allow correlation in errors as long as observations share at least one cluster. To be precise, when we are clustering over dimensions G and H with GH denoting the intersection, the CGM estimator is

$\hat{V}_{CGM} = \hat{V}_{LZG} + \hat{V}_{LZH} - \hat{V}_{LZGH}$. Menzel (2021) points out that asymptotic normality fails in some pathological cases, a situation unique to multiway clustering with no analog in one-way clustering, and proposes a bootstrap procedure that is adaptive. MacKinnon et al. (2021) study regression models with clustering in two dimensions, provide results for validity of the CGM estimator and compare the CGM estimator with an alternative proposed by Davezies et al. (2018).

The most relevant work from the design-based literature is Abadie et al. (2017) (henceforth AAIW), who consider the appropriateness of oneway clustering. They argue clustering is either a sampling design or experimental design issue, and give conditions where clustering is appropriate. Sampling uncertainty comes from the fact that we only observe a subset of the population. Assignment uncertainty comes from how we only observe one possible state of the world. In the Rubin causal model with binary treatment, we only observe either $Y(1)$ or $Y(0)$ and not both. AAIW find that clustering is appropriate when sampling follows a two-stage process: in stage 1, a subset of clusters were sampled randomly from a population of clusters and in stage 2, units sampled randomly from sampled clusters. Clustering is appropriate when assignment is correlated within clusters. Their results are specific to clustering on a single dimension, which requires assignment and sampling to be independent over clusters. But we have already established that there are many empirical examples where such an assumption is not realistic. It is also not obvious how their findings generalize to multiway clustering. Further, unlike this paper, they do not have asymptotic results.

The main challenge of moving from one-way to multiway clustering is in the asymptotic behavior. Consider simplest mean problem: $\hat{\mu} = \frac{1}{N}\sum_{i=1}^{N} W_i$. For law of large numbers (LLN), we want iid observations, but in clustering context, we cannot guarantee that. With one-way clustering, $\hat{\mu} = \sum_{g=1}^{G} \frac{N_g}{N}\left(\sum_{i \in S_g} W_i\right)$, then use some LLN for weighted averages, since $\left(\sum_{i \in S_g} W_i\right)$ are independent over g's. But with two-way clustering, that trick no longer works because we lose independence over clusters.

The current solution for asymptotics in multiway clustering is to apply the Aldous-Hoover representation, and we have asymptotic guarantees in Davezies et al. (2018) when separate exchangeability holds. This is the assumption that $(\mathcal{S}_{gh}) \overset{d}{=} (\mathcal{S}_{\pi_1(g)\pi_2(h)})$, where $(\mathcal{S}_{gh})$ is the distribution of (W, ε) over clusters g, h, and π's permutate the vectors. To illustrate this with an example, if we are interested in the effect of minimum wages on employment, this is saying that if we had treated Pennsylvania instead of New Jersey, we would obtain the same distribution of outcomes. However,

by construction, this is violated in a design-based framework because $\varepsilon_i = W_i\varepsilon_i(1) + (1 - W_i)\varepsilon_i(0)$ depends on nonstochastic $\varepsilon_i(w)$. The distribution cannot be the same when the W_i's are weighted differently.

To overcome these challenges, this paper formalizes conditions to rule out those pathological cases of non-normality in Menzel (2021). I apply the Aldous-Hoover representation as before to generate terms that are independent over clusters that feature additively, allowing the application of LLN, even without separate exchangeability.

1.2 Empirical Implications

Using the main takeaway that only the assignment mechanism is a relevant consideration for clustering, we can revisit applied work that used a model-based justification for clustering, and consider what they could have done differently if they were to use the preferred design-based approach and are interested in the sample ATE. There are some situations where papers clustered on dimensions that they did not have to, and situations where papers did not cluster on dimensions that they should. I offer three examples of how the finding plays out.

Dube et al. (2010) want to find the effect of minimum wage on employment, and cluster by state and border segment. They observe 504 border counties out of 1139, which is non-negligible, which we may treat as being randomly chosen. Minimum wage assignment occurs at the state level instead of the border segment, so it is sufficient to cluster on the state level.

Michalopoulos and Papaioannou (2013) study the effect of pre-colonial institutions on modern development. They clustered by modern country and ethnolinguistic family. But pre-colonial institutions are more likely determined by pre-colonial regions than the modern country, so they should cluster on pre-colonial regions instead of the modern country. The latter would be justified in the model-based framework, but the former in the design-based framework.

Nunn and Wantchekon (2011) want to find the effect of slave trade on trust today. They cluster by ethnic groups and district. Unit of observation is an individual. Both these dimensions affect assignment because people of similar ethnicities have similar history of slave trade. District is relevant since people in the same location are more likely hit by slave trade.

2 A Simple Illustration

The main findings in this paper can be numerically illustrated with a monte carlo simulation. If we cluster on more dimensions than we have to, results are conservative, and this simulation gives a sense of how conservative this can be. Namely, in the data-generating process (DGP) I used, clustering on too many dimensions gives standard errors 13 times larger than what is required to get correct coverage.

There are $C_G = 200$ clusters on the G dimension and $C_H = 200$ clusters on the H dimension. I draw a sample of $N = 100000$ observations, with an individual error term of $\varepsilon_i \sim N(0,1)$. Each i is assigned randomly assigned (with equal probability over all 200 discrete values) a g and h value on the respective clustering dimensions. Then, allow the treatment effect to be entirely determined by:

$$\tau_i = \tau_{G_i} + \tau_{H_i}$$

$$\tau_g = \begin{cases} -2 & \text{w.p. } 1/2 \\ +2 & \text{w.p. } 1/2 \end{cases}$$

$$\tau_h = \begin{cases} -0.5 & \text{w.p. } 1/2 \\ +0.5 & \text{w.p. } 1/2 \end{cases}$$

I use four different designs to assign treatment status W. In all cases, set $\mu := E[W_i] = .5$. In the first design ("neither"), W_i is assigned iid over all individuals, so $p(W_i = 1) = 1 - p(W_i = 0) = .5$. In the second design ("G"), for each g, I draw $q_g \sim U[0,1]$. Every individual in cluster g is independently assigned treatment with probability q_g. The third design ("H") is identical to that of g, except that I use $q_h \sim U[0,1]$ instead. The four design ("both") has assignment dependent on both dimensions: draw q_g and q_h independently from $U[0,1]$. Assign treatment to observation i if both q_{G_i} and q_{H_i} are less than $1/\sqrt{2}$, i.e., $W_i = 1[q_{G_i} < 1/\sqrt{2}, q_{H_i} < 1/\sqrt{2}]$.

	Neither	G	H	Both
EHW_se	0.011	0.011	0.011	0.011
LZG_se	0.142	0.163	0.142	0.169
LZH_se	0.037	0.037	0.042	0.043
CGM_se	0.146	0.167	0.147	0.173
LZGH_se	0.015	0.016	0.016	0.018
EHW_cov_sam	0.983	0.208	0.670	0.164
LZG_cov_sam	1.000	1.000	1.000	1.000
LZH_cov_sam	1.000	0.621	1.000	0.641
CGM_cov_sam	1.000	1.000	1.000	1.000
LZGH_cov_sam	0.999	0.298	0.855	0.288
EHW_cov_pop	0.128	0.100	0.126	0.104
LZG_cov_pop	0.943	0.945	0.947	0.934
LZH_cov_pop	0.387	0.314	0.418	0.396
CGM_cov_pop	0.950	0.950	0.954	0.941
LZGH_cov_pop	0.176	0.140	0.177	0.166

Table 1: Monte Carlo Results. With $N = 100000$ observations, $C_G = C_H = 200$ clusters and $nsim = 2000$ simulations. Columns denote the assignment mechanism, representing whether assignment is determined iid (neither), or determined on the G, H or both dimensions. Coverage denotes the fraction of simulations whose 95% confidence interval (constructed by $\hat{\tau} \pm 1.96se$) covers the correct object, which is the sample average treatment effect (ATE), or the population ATE (which is 0 by construction). Rows denote the type of clustering used to perform inference, and the mean standard errors, coverage for the sample ATE, and coverage for the population ATE respectively.

Outcomes are determined by:

$$Y_i = \tau_i W_i + \varepsilon_i$$

With each data draw, I regress Y on W and calculate standard errors using 5 different approaches. We can use the heteroskedasticity-robust standard erorrs (EHW), clustering on either the G or H dimension (LZG, LZH), cluster on the intersection of the two dimensions (i.e., every $G \cap H$ is treated as a separate cluster, denoted LZGH), or use twoway clustering given by CGM. The standard errors and coverage is recorded over $nsim = 2000$ simulations, and reported in Table 1. By construction, the average treatment effect (ATE) for the population is $E[\tau_i] = 0$, and the sample ATE is $\frac{1}{N}\sum_{i=1}^{N} \tau_i$.

Observe that the DGP used has cluster-dependent error on G and H. We can write $\tau_i = \tau +$

$\tilde{\tau}_{G_i} + \tilde{\tau}_{H_i}$, so τ is the population ATE and $\tilde{\tau}_{G_i}, \tilde{\tau}_{H_i}$ will be part of the unobserved error term. In the model-based framework, there is clearly a need to cluster on both the G and H dimensions to get correct coverage, but this simulation suggests otherwise in the design-based framework, which should change the way we think about the Moulton (1986) problem.

Consider the assignment design where W_i is iid over all observations ("neither"). The CGM standard error is $0.142/0.011 \approx 13$ times larger than the EHW standard error. But when we look at the coverage for the sample ATE, the EHW coverage is still correct as it exceeds 0.95, but CGM has 100% coverage. This shows how CGM is too conservative, and we could have reduced standard errors by a factor of 13 while having correct coverage properties for the sample ATE. In contrast, the EHW errors under-cover the population ATE severely. This occurs due to clustered sampling behavior as we only observed a negligibly small fraction of clusters from the superpopulation. If we were interested in the population ATE, we would have to cluster on G and H, and CGM gives correct coverage asymptotically. These observation highlights the differences between the model-based and design-based frameworks. In a model-based world, we would be worried about the cluster-dependent error to get correct coverage, but not so in the design-based world.

Next, consider the G column, where assignment occurs on the G dimension. If we had chosen to cluster on the G dimension, we would have coverage of at least 95% for the sample ATE. But if we used EHW or LZH, there will be severe undercoverage for the sample ATE. This emphasizes the fact that clustered behavior on the assignment dimension matters for inference. In the H column, it is sufficient to cluster on the H dimension for the sample ATE, since assignment occurs on the H dimension. If we had chosen to do CGM instead, standard errors will be 3 times larger than what is actually required. Finally, when assignment occurs on both dimensions, CGM gives correct coverage.

One may observe that using LZG appears to give 100% coverage regardless of the assignment process. This is a result of the DGP, where the heterogeneity in treatment effect generated on the G dimension is much larger than the heterogeneity generated on the H dimension.

The rest of this paper formally proves results that variance estimators that cluster only on assignment dimensions give correct coverage, giving theoretical guarantees for these monte carlo observations.

3 Setting

We are interested in the average treatment effect (ATE) of W on Y for the sample. Assignment W is treated as random. Denote potential outcome Y for individual i with treatment status w as $Y_i(w)$. This setting implicitly assumes the Stable Unit Treatment Value Assumption (SUTVA) in that the potential outcomes are not affected by treatment status of other individuals. Defining $\bar{Y}(w) := \frac{1}{N}\sum_{i=1}^{N} Y_i(w)$, with sample size N and binary treatment, the object of interest is written as:

$$\tau := \frac{1}{N}\sum_{i=1}^{N}(Y_i(1) - Y_i(0)) = \bar{Y}(1) - \bar{Y}(0) \tag{1}$$

Define the individuals difference from the mean potential outcome as:

$$\varepsilon_i(w) := Y_i(w) - \frac{1}{N}\sum_{i=1}^{N} Y_i(w) = Y_i(w) - \bar{Y}(w) \tag{2}$$

This is a deterministic function of potential outcomes, so conditioning on potential outcomes is equivalent to conditioning on the $\varepsilon_i(w)$ objects. Hence, by construction, $\frac{1}{N}\sum_{i=1}^{N}\varepsilon_i(w) = 0$ for all w, and the $\varepsilon_i(w)$ objects are nonstochastic functions of potential outcomes that we condition on. This setup allows for heterogeneous treatment effects, and the assumption of homogeneous treatment effects is equivalent to $\varepsilon_i(1) - \varepsilon_i(0) = \tau$ for all i.[1]

Remark 1. *Using more familiar notation, the setup can equivalently be written as a linear model with $Y_i = \alpha + \tau W_i + \varepsilon_i$. Then, α is the average potential outcome for non-treatment for the sample $\alpha := \frac{1}{N}\sum_{i=1}^{N} Y_i(0)$.*

This also implies that the randomness in Y_i and ε_i can only be attributed to randomness in W_i.

$$Y_i = W_i Y_i(1) + (1 - W_i)Y_i(0)$$
$$\varepsilon_i = W_i \varepsilon_i(1) + (1 - W_i)\varepsilon_i(0)$$

Consider a general clustered assignment setup where assignment is correlated as long as units of observation share at least one cluster. Individual i is in cluster G_i on the G dimension and in cluster H_i on the H dimension, with an individual unobservable v_i.

[1] There is an appropriate analog for the population ATE, but that is beyond the scope of this paper.

Assumption 1. *(Assignment). Binary W_i and W_j are independent whenever i and j do not share any cluster. Further, W is separately exchangeable (i.e., for $W_{gh} = (W_i)_{i \in S_{gh}}$, $(W_{gh})_{g \geq 1, h \geq 1} \stackrel{d}{=} (W_{\pi_1(g)\pi_2(h)})_{g \geq 1, h \geq 1}$ where π are permutation functions) and $\mu := E[W_i] \in (0,1)$.*

Remark 2. *An equivalent representation of the assumption can be stated as follows: A_g is a random variable drawn iid over clusters on the G dimension; B_h is a random variable drawn iid over clusters on the H dimension; v_i is drawn iid over individuals. A_g, B_h, and v_i are mutually independent. For observation i, binary assignment is given by $W_i := f(A_{G_i}, B_{H_i}, v_i) \in \{0,1\}$, and $\mu := E[W_i] \in (0,1)$. This equivalence is the result of Lemma C1 of Davezies et al. (2018).*

This assumption is equivalent to the combination of Assumptions 4 and 6 in MacKinnon et al. (2021). Further, this assumption rules out edge cases $\mu \in \{0,1\}$ where either everyone is treated or everyone is untreated. If that is so, we cannot hope to make inferences about the causal effect. A further substantial implication of this assumption is that there is no selection into treatment based on outcome.

Let N_g^G denote the number of units sampled that is in cluster g on the G dimension and N_h^H the number of units sampled from cluster h on the H dimension. Similarly, let M_N^G denote the number of clusters on the G dimension in sample size N, with an analogous definition for M_N^H. Let S_{gh} denote the set of individuals that are in cluster g on the G dimension and cluster h on the H dimension. Similarly, N_{gh} denotes the number of units sampled in the intersection of cluster g and cluster h. Define cluster specific residuals as $\tilde{\varepsilon}_{gh}(w) := \frac{1}{N_{gh}} \sum_{i \in S_{gh}} \varepsilon_i(w)$.

Assumption 2. *The following hold:*

(i) $\lim_{N \to \infty} (M_N^G)^{-1} = 0$ *and* $\lim_{N \to \infty} (M_N^H)^{-1} = 0$.

(ii) $\max_g \frac{N_g^G}{N} \to 0$ *and* $\max_h \frac{N_h^H}{N} \to 0$.

(iii) Potential outcomes are bounded i.e., $\max_{i,w} |Y_i(w)| < \infty$.

Assumption 2(i) requires the number of clusters to get large on both dimensions, thereby ruling out finite M_N^G or M_N^H. This is required for consistency because we are getting independent draws of A_g and B_h. If only one of the clustering dimensions gets large, say G, and we have an $f(.)$ where B_h features additively, then the mean of B_h and hence of W_i does not converge asymptotically.

Assumption 2(ii) rules out disproportionately large clusters. A possible violation is where we have one unit in every cluster $g \geq 2$, and every remaining observation in the first cluster. Then, $\sum_g (N_g^G)^2 \approx N^2$, so $\sum_g \left(\frac{N_g^G}{N}\right)^2 \approx 1$, and does not converge to zero. This condition is the same as Assumption 1 of Hansen and Lee (2019).[2] The assumption is equivalent to $\sum_g \left(\frac{N_g^G}{N}\right)^2 \to 0$ and $\sum_h \left(\frac{N_h^H}{N}\right)^2 \to 0$. To see this,

$$\frac{\sum_g (N_g^G)^2}{N^2} \leq \frac{\left(\sum_g N_g^G\right) \max_g N_g^G}{N^2} = \frac{\max_g N_g^G}{N} \to 0$$

For the other direction,

$$\max_g \frac{N_g^G}{N} = \left(\max_g \frac{(N_g^G)^2}{N^2}\right)^{1/2} \leq \left(\sum_g \frac{(N_g^G)^2}{N^2}\right)^{1/2} \to 0$$

Assumption 2(iii) is comparable to existing literature, which often imposes a moment condition on $E[Y_i^k]$.[3] Since $Y_i = W_i Y_i(1) + (1 - W_i) Y_i(0)$ in this context, with W_i discrete, for expectation of any moment of Y_i to be bounded, we require potential outcomes to be bounded. Abadie et al. (2017) similarly imposes bounded potential outcomes.

Remark 3. *Assumption 2(ii) and Assumption 2(iii) are stronger but more interpretable than what we actually need. Based on the current proof strategy, in Assumption 2(ii), it is sufficient that* $\sum_g \left(\frac{N_g^G}{N}\right)^{1+\delta} \to 0$ *and* $\sum_h \left(\frac{N_h^H}{N}\right)^{1+\delta} \to 0$. *Here,* $\delta = 1$ *is a special case of this condition. In Assumption 2(iii), the condition required is that* $\sum_g \frac{(N_g^G)^2}{N^2} (\tilde{\varepsilon}_g^w)^2 \to 0$ *and* $\sum_h \frac{(N_h^H)^2}{N^2} (\tilde{\varepsilon}_h^w)^2 \to 0$, *with* $\tilde{\varepsilon}_g^w$ *defined in Equation 6. Bounded outcomes ensures that this holds. When using the weaker version of Assumption 2(iii), simply use the Markov inequality instead of LLN whenever Assumption 2(iii) is used in the proofs.*

A canonical sample that satisfies these conditions is where $N_{gh} = 1$ for all g, h, and $M_N^G = M_N^H$ goes to infinity. An example of this is a panel dataset with a large number of individuals and large number of time periods. Then, $\frac{N_g^G}{N} = \frac{1}{M_N^H}$, so $\sum_g \left(\frac{N_g^G}{N}\right)^2 = \frac{M_N^G}{(M_N^H)^2} = \frac{1}{M_N^H} \to 0$.

Assumptions 1 and 2 are maintained throughout, and are sufficient for consistency of the OLS estimator. To obtain consistency of variance estimators and asymptotic normality, we require

[2] This is also identical to Assumption 2(ii) of Adao et al. (2019).
[3] See, for instance, regularity conditions in Xu (2021).

additional assumptions that will be presented in the next section.

4 Results

4.1 Consistency of Estimator

To obtain consistency of the OLS estimator, I first prove a weak law of large numbers (WLLN) with weights that will be helpful for derivations later. As a special case, the weights a_i sum to 1, which is the case when we are interested in for the sample object.

Lemma 1. *(WLLN for Weighted Sum). Suppose a_n is a vector of weights of length n and a_{ni} denotes the ith element of a_n. Random variables X_i are distributed with bounded mean μ_i and are mean independent (i.e., for all $i \neq j$, $E[X_i|X_j] = \mu_i$). If $E|X_i|^{1+\delta} < \infty$ for some $\delta \in (0, 1]$, and nonstochastic sequence of $\{a_n\}$ satisfies $\sum_{i=1}^n a_{ni}^{1+\delta} \to 0$ as $n \to \infty$, then $\sum_{i=1}^n a_{ni}(X_i - \mu_i) \xrightarrow{p} 0$.*

The vanilla WLLN is a special case of this with weights $a_{in} = 1/n$ for all i. The sum of weights raised to $1+\delta$ converging to zero rules out having a finite number of nonzero weights. For instance, $a_{ni} = 1$ for $i = 1$ and $a_{ni} = 0$ otherwise violates the condition, because only the first observation is used in the estimator, and we cannot hope to have a consistent estimate. The proof of the result applies the von Bahr and Esseen (1965) inequality, which requires a martingale difference sequence (implied by mean independence) and finite $1 + \delta$ moments.

Due to Assumption 2(ii), we already have $\sum_i a_{in}^2 \to 0$ in the relevant cases. This implies that $\sum_i a_{in}^{1+\delta} \to 0$ for all $\delta \in (0, 1]$. To see this, $\sum_i a_{in}^{1+\delta} \leq \sum_i a_{in}^2 \to 0$ since all the weights are in $[0, 1]$. Consequently, as long as $1 + \delta$ moments exist for the random variable, the LLN can be applied.

Define estimator:

$$\begin{aligned}\hat{\tau} &:= \frac{1}{\sum_{i=1}^N W_i} \sum_{i=1}^N W_i Y_i - \frac{1}{N - \sum_{i=1}^N W_i} \sum_{i=1}^N (1 - W_i) Y_i \\ &= \frac{\frac{1}{N}\sum_{i=1}^N W_i Y_i(1)}{\frac{1}{N}\sum_{i=1}^N W_i} - \frac{\frac{1}{N}\sum_{i=1}^N (1 - W_i) Y_i(0)}{1 - \frac{1}{N}\sum_{i=1}^N W_i}\end{aligned}$$

To simplify the estimator, it is helpful to have the mean $\bar{W} = \frac{1}{N}\sum_{i=1}^N W_i \xrightarrow{p} \mu$.

Lemma 2. *If Assumptions 1 and 2(i)-2(ii) hold, then* $\frac{1}{N}\sum_{i=1}^{N} W_i - \mu = o_P(1)$.

Showing that $\bar{W}$ converges is nontrivial due to the dependence structure from Assumption 1. To address this, use the representation attributed to Aldous (1981) and Hoover (1979) to generate weighted sums of independent draws, allowing the application of Lemma 1. Notably, W_i is still separately exchangeable, so existing results follow neatly. Assumption 1 gives the mean independence required and Assumption 2 allows asymptotic properties to be well-behaved to apply LLN. We did not require Assumption 2(iii) because it is irrelevant to W. The random variable is already bounded by making it binary from Assumption 1.

To be precise, consider these conditional expectations that is helpful in later exposition:

$$\begin{aligned} Q_g^G &:= E_{h,v}[f(A_g, B_h, v)] \\ Q_h^H &:= E_{g,v}[f(A_g, B_h, v)] \\ Q_{gh}^{GH} &:= E_v[f(A_g, B_h, v)] \end{aligned} \tag{3}$$

Here, $E_{h,v}$ denotes integration over the distribution of (B_h, v). This is defined on the distribution of W_i and not the observed sample. Since (A_g, B_h, v) are mutually independent, Q_g^G is also independent of (B_h, v) draws: it is merely a transformation of the A_g random variable. Since A_g is drawn iid, Q_g^G is also iid, with mean μ. The same reasoning applies to the other two objects.

With this result, we can make an approximation so that we can focus on weighted sums.

Lemma 3. *If Assumptions 1, 2(i)-2(ii) hold, then, with N sampled units, and* $\eta_i := \frac{(W_i-\mu)\varepsilon}{\sqrt{\mu(1-\mu)}}$,

$$(\hat{\tau} - \tau) = \frac{1}{N\sqrt{\mu(1-\mu)}} \sum_{i=1}^{N} \eta_i + o_P(1) \tag{4}$$

The expression derived here generalizes the approximation made in Abadie et al. (2017): setting $\mu = 1/2$ gives a result identical to their approximation. Lemma 3 follows from convergence in Lemma 2, some algebra, and continuous mapping theorem (CMT).

This approximation allows us to focus on object $\frac{1}{N\sqrt{\mu(1-\mu)}} \sum \eta_i$. Now, we turn to the main result

of this subsection: the OLS estimator is consistent.

Theorem 1. *Under Assumptions 1 and 2,* $\hat{\tau} - \tau = o_P(1)$.

The proof strategy for this theorem is similar to the previous lemma, by using the Aldous-Hoover representation. Due to the approximation in Lemma 3, it is sufficient to show $\frac{1}{N}\sum_{i=1}^{N} \eta_i \xrightarrow{p} 0$. In contrast to W_i, η_i is not separately exchangeable, due to our conditioning on potential outcomes.

Apply the Aldous-Hoover representation and use the notation in Equation 3. Since $E[\eta_i] = \sqrt{\mu(1-\mu)}(\varepsilon_i(1) - \varepsilon_i(0))$,

$$\begin{aligned}
\eta_i &= \sqrt{\mu(1-\mu)}(\varepsilon_i(1) - \varepsilon_i(0)) + a_{G_i} + b_{H_i} + c_{G_iH_i} + e_i \\
a_{G_i} &= E[\eta_i | A_{G_i}] - \sqrt{\mu(1-\mu)}(\varepsilon_i(1) - \varepsilon_i(0)) \\
b_{H_i} &= E[\eta_i | B_{H_i}] - \sqrt{\mu(1-\mu)}(\varepsilon_i(1) - \varepsilon_i(0)) \\
c_{G_iH_i} &= E[\eta_i | A_{G_i}, B_{H_i}] - E[\eta_i | A_{G_i}] - E[\eta_i | B_{H_i}] + \sqrt{\mu(1-\mu)}(\varepsilon_i(1) - \varepsilon_i(0)) \\
e_i &= \eta_i - E[\eta_i | A_{G_i}, B_{H_i}]
\end{aligned}$$

Then, each of these objects has mean zero. Further, we want to aggregate the a_{G_i} within each cluster so that we get independence over clusters to get convergence of the a term, and a similar method is applied over b_{H_i}. Independence of B_h draws conditional on A_g allows convergence of the c term.

With consistency of the OLS estimator in the design-based framework, we can now turn to its variance, and the performance of feasible variance estimators relative to the true variance, the subject of the next subsection.

4.2 Variance Estimators

The LZ variance estimator on the G dimension is:

$$\hat{V}_{LZG} := \frac{1}{N^2} \sum_g \left(\sum_i 1[G_i = g] \frac{(W_i - \hat{\mu})\hat{\varepsilon}_i}{\hat{\mu}(1-\hat{\mu})} \right) \left(\sum_j 1[G_j = g] \frac{(W_j - \hat{\mu})\hat{\varepsilon}_j}{\hat{\mu}(1-\hat{\mu})} \right) \tag{5}$$

Let's use the true μ, ε instead of the $\hat{\mu}$ and $\hat{\varepsilon}$:

$$V_{LZG} := \frac{1}{N^2} \sum_g \left(\sum_i 1[G_i = g] \frac{(W_i - \mu)\varepsilon_i}{\mu(1-\mu)} \right) \left(\sum_j 1[G_j = g] \frac{(W_j - \mu)\varepsilon_j}{\mu(1-\mu)} \right)$$

Consider its expectation:

$$\mathbb{V}_{LZG} := E[V_{LZG}]$$

Define variance estimators for EHW, LZH and CGM similarly. The objective of this section is to find $Var(\hat{\tau})$, and compare it against EHW, LZ and CGM variance estimators. Note that we cannot use the expectation of $\hat{V}$ directly because it is not guaranteed to exist when dividing by $\hat{\mu}$.

By conducting the same exercise of using the true μ, ε instead of $\hat{\mu}$ and $\hat{\varepsilon}$ in the expression for $\hat{\tau} - \tau$, we can similar consider the variance of $\frac{1}{N\sqrt{\mu(1-\mu)}} \sum_{i=1}^N \eta_i$. Hence, define

$$\mathbb{V}_\eta := Var\left(\frac{1}{N\sqrt{\mu(1-\mu)}} \sum_{i=1}^N \eta_i \right)$$

If we can make the variance estimators $\hat{V}$ converge to their respective $\mathbb{V}$ asymptotically, then it remains to compare the $\mathbb{V}$ objects. To save on notation, define:

$$\begin{aligned}
\nu_i &:= E[\eta_i] = \sqrt{\mu(1-\mu)}(\varepsilon_i(1) - \varepsilon_i(0)) \\
\tilde{\nu}_g &:= \frac{1}{N_g^G} \sum_{i \in S_g} \sqrt{\mu(1-\mu)}(\tilde{\varepsilon}_g(1) - \tilde{\varepsilon}_g(0)) \\
\varepsilon_i^w &:= \left(\sqrt{\frac{1-\mu}{\mu}} \varepsilon_i(1) + \sqrt{\frac{\mu}{1-\mu}} \varepsilon_i(0) \right) \\
\tilde{\varepsilon}_g^w &:= \left(\sqrt{\frac{1-\mu}{\mu}} \tilde{\varepsilon}_g(1) + \sqrt{\frac{\mu}{1-\mu}} \tilde{\varepsilon}_g(0) \right)
\end{aligned} \tag{6}$$

These are nonstochastic objects. The latter two objects are defined in that way because the Aldous-

Hoover representation simplifies to:

$$\begin{aligned}
\eta_i &= \nu_i + a_{G_i} + b_{H_i} + c_{G_iH_i} + e_i \\
a_{G_i} &= (Q^G_{G_i} - \mu)\varepsilon^w_i \\
b_{H_i} &:= (Q^H_{H_i} - \mu)\varepsilon^w_i \\
c_{G_iH_i} &:= (Q^{GH}_{G_iH_i} - Q^G_{G_i} - Q^H_{H_i} + \mu)\varepsilon^w_i
\end{aligned}$$

Further, define variances as:

$$\begin{aligned}
V_G &:= E[(Q^G_g - \mu)^2] \\
V_H &:= E[(Q^H_h - \mu)^2] \\
V_{GH} &:= E[(Q^{GH}_{gh} - \mu)^2]
\end{aligned}$$

The assumptions we have thus far are not strong enough to give convergence of variances, so we introduce another.

Assumption 3. *For all g,* $\max_h \frac{N_{gh}}{N^G_g} \to 0$ *and for all h,* $\max_g \frac{N_{gh}}{N^H_h} \to 0$.

Assumption 3 plays a role similar to Assumption 2(ii), just that it occurs at every intersection. For instance, if we are clustering by state and industry, it rules out the behavior where we only have oil firms in Texas, because subsequent additions to the series will only be zeroes. This assumption necessarily implies we must allow $N^H_h \to \infty$. If we have a fixed number of observations per cluster (i.e., N^H_h fixed), Assumption 3 is violated because convergence cannot occur even when N gets large. We did not require the assumption before because these interactive terms only appear when using the variance estimator. The convergence of the sum of squares allows application of Lemma 1 yielding the consistency of the variance estimators. Consider the canonical example of $N_{gh} = 1$ again. Assumption 3 is satisfied because $\sum_h \left(\frac{N_{gh}}{N^G_g}\right)^2 = \sum_h \frac{1}{(M^H_N)^2} = \frac{1}{M^H_N} \to 0$. As before, if we have separately exchangeable cluster sizes, as in most of the literature in multiway clustering, we get Assumption 3 for free. There is no appropriate analog in the one-way clustering setup because the requirement for interactive terms to converge is unique to multiway clustering.

In a weaker setup without bounded outcomes, we would additionally require the existence of $2 + \delta$

moments of the outcome. Since we have already assumed bounded potential outcomes, no further assumption on the outcome is required.

The $E[\eta_i^2]$ term gives us the limit of the EHW estimator. The variance expressions are obtained through tedious algebra detailed in the appendix. Convergence is proved by application of the strategies already discussed to allow the application of LLN. Assumption 3 features due to the need to split cross products. For instance,

$$\begin{aligned}\frac{1}{N}\sum_{g,h} N_{gh}^2(Q_g^G-\mu)(Q_h^H-\mu)(\tilde{\varepsilon}_{gh}^w)^2 &= \frac{1}{N}\sum_g (Q_g^G-\mu)\sum_h N_{gh}^2(Q_h^H-\mu)\\ &= \frac{1}{N}\sum_g (N_g^G)(Q_g^G-\mu)\sum_h \frac{N_{gh}^2}{(N_g^G)}(Q_h^H-\mu) = o_P(1)\end{aligned}$$

Theorem 2. *Under Assumptions 1, 2 and 3, for $t \in \{EHW, LZG, LZH, CGM\}$, either $\mathbb{V}_t = 0$ or*

$$\mathbb{V}_t^{-1}\hat{V}_t = 1 + o_P(1)$$

Similarly, either $\mathbb{V}_\eta = 0$ or

$$\mathbb{V}_\eta^{-1} Var(\hat{\tau}) = 1 + o_P(1)$$

where

$$\begin{aligned}\mathbb{V}_{EHW} &= \frac{1}{N^2\mu(1-\mu)}\sum_{i=1}^N (1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2\\ \mathbb{V}_{LZG} &= \frac{1}{N^2\mu(1-\mu)}\{\sum_g (N_g^G)^2(\tilde{\nu}_g^2 + V_G(\tilde{\varepsilon}_g^w)^2) + \sum_{g,h}(N_{gh}^2)[(\tilde{\nu}_{gh}^2 + V_{GH}(\tilde{\varepsilon}_{gh}^w)^2) - (\tilde{\nu}_g^2 + V_G(\tilde{\varepsilon}_g^w)^2)]\\ &\quad + \sum_i [(1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2 - \nu_i^2 - V_{GH}(\varepsilon_i^w)^2]\}\\ \mathbb{V}_{CGM} &= \frac{1}{N^2\mu(1-\mu)}\{\sum_g (N_g^G)^2(\tilde{\nu}_g^2 + V_G(\tilde{\varepsilon}_g^w)^2) + \sum_h (N_h^H)^2(\tilde{\nu}_h^2 + V_H(\tilde{\varepsilon}_h^w)^2)\\ &\quad + \sum_{g,h}(N_{gh}^2)[(\tilde{\nu}_{gh}^2 + V_{GH}(\tilde{\varepsilon}_{gh}^w)^2) - (\tilde{\nu}_g^2 + V_G(\tilde{\varepsilon}_g^w)^2) - (\tilde{\nu}_h^2 + V_H(\tilde{\varepsilon}_h^w)^2)]\\ &\quad + \sum_i [(1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2 - \nu_i^2 - V_{GH}(\varepsilon_i^w)^2]\}\end{aligned}$$

$$\mathbb{V}_\eta = \frac{1}{N^2\mu(1-\mu)}\{\sum_g (N_g^G)^2 V_G(\tilde{\varepsilon}_g^w)^2 + \sum_h (N_h^H)^2 V_H(\tilde{\varepsilon}_h^w)^2$$
$$+ \sum_{g,h}(N_{gh}^2)[V_{GH}(\tilde{\varepsilon}_{gh}^w)^2 - V_G(\tilde{\varepsilon}_g^w)^2 - V_H(\tilde{\varepsilon}_h^w)^2] + \sum_i [(1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2 - \nu_i^2 - V_{GH}(\varepsilon_i^w)^2]\}$$

Then, we can compare the respective variances to the true variance:

Corollary 1. *The difference between asymptotic variances and target variance is:*

$$N^2\mu(1-\mu)(\mathbb{V}_{CGM} - \mathbb{V}_\eta)$$
$$= \sum_g \left((N_g^G)^2 - \sum_h N_{gh}^2\right)(\tilde{\nu}_g^2) + \sum_h \left((N_h^H)^2 - \sum_g N_{gh}^2\right)(\tilde{\nu}_h^2) + \sum_{g,h}(N_{gh}^2)(\tilde{\nu}_{gh}^2) \geq 0$$
$$N^2\mu(1-\mu)(\mathbb{V}_{LZG} - \mathbb{V}_\eta)$$
$$= \sum_g (N_g^G)^2(\tilde{\nu}_g^2) - \sum_h (N_h^H)^2 V_H(\tilde{\varepsilon}_h^w)^2 + \sum_{g,h}(N_{gh}^2)[\tilde{\nu}_{gh}^2 - \tilde{\nu}_g^2 + V_H(\tilde{\varepsilon}_h^w)^2]$$
$$N^2\mu(1-\mu)(\mathbb{V}_{EHW} - \mathbb{V}_\eta)$$
$$= -\sum_g (N_g^G)^2 V_G(\tilde{\varepsilon}_g^w)^2 - \sum_h (N_h^H)^2 V_H(\tilde{\varepsilon}_h^w)^2 - \sum_{g,h}(N_{gh}^2)[V_{GH}(\tilde{\varepsilon}_{gh}^w)^2 - V_G(\tilde{\varepsilon}_g^w)^2 - V_H(\tilde{\varepsilon}_h^w)^2]$$
$$+ \sum_i [\nu_i^2 + V_{GH}(\varepsilon_i^w)^2]$$

There are some special cases to consider. If both clustering dimensions are identical, then $(N_g^G)^2 - \sum_h N_{gh}^2 = 0$ and $(N_h^H)^2 - \sum_g N_{gh}^2 = 0$, so the expressions for $\mathbb{V}_{CGM} - \mathbb{V}_\eta$ and $\mathbb{V}_{LZG} - \mathbb{V}_\eta$ are identical. If H clusters are completely nested within G clusters, then $(N_h^H)^2 - \sum_g N_{gh}^2 = 0$, and again expressions for $\mathbb{V}_{CGM} - \mathbb{V}_\eta$ and $\mathbb{V}_{LZG} - \mathbb{V}_\eta$ are identical. As intuitively expected, these two cases collapse to the one-way clustering results, where we use the larger cluster.

Clearly, $(N_g^G)^2 - \sum_h N_{gh}^2 \geq 0$, so the CGM estimator is conservative. It will be correct when $\tilde{\nu}_{gh} = 0$, which occurs when there is constant treatment effect.

Considering the result for $\mathbb{V}_{LZG} - \mathbb{V}_\eta$, if there is no assignment variation on H, then $V_H = 0$ and the expression reduces to $\sum_g (N_g^G)^2(\tilde{\nu}_g^2) + \sum_{g,h}(N_{gh}^2)[\tilde{\nu}_{gh}^2 - \tilde{\nu}_g^2] \geq 0$. However, if there is indeed variation by assignment on H and the researcher clusters on G, we can get a result where $\mathbb{V}_{LZG} - \mathbb{V}_\eta < 0$ due to negative $\sum_h (N_h^H)^2 V_H(\tilde{\varepsilon}_h^w)^2$, so using that variance estimator will under-cover. This problem

does not go away even if we have constant treatment effects such that $\tilde{\nu} = 0$. The result for the EHW estimator has a similar interpretation.

4.3 Asymptotic Normality

The objective of this subsection is to show that the OLS estimator is asymptotically normal. This requires an additional assumption, and we want to write the object in a form amenable to using the Lindeberg Central Limit Theorem (CLT). First, define the sum of variances as the following:

$$
\begin{aligned}
s_N &:= \max\{s_N^G, s_N^H\} \\
(s_N^G)^2 &:= \sum_g V_G\left(\tilde{\varepsilon}_g^w\right)^2 \left(\frac{N_g^G}{N}\right)^2 \\
(s_N^H)^2 &:= \sum_h V_H\left(\tilde{\varepsilon}_h^w\right)^2 \left(\frac{N_h^H}{N}\right)^2
\end{aligned}
$$

To motivate the assumption required in the proof strategy, first consider two examples from Menzel (2021) - the first of which is not asymptotically normal and the second is.

Example 1. *Suppose α_i, γ_t, and ε_{it} are iid mean zero. We have data generated from $y_{it} = \alpha_i\gamma_t + \varepsilon_{it}$, and have one observation per (i,t) pair. We are interested in the mean of Y.*

$$
\begin{aligned}
\frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\sum_{t=1}^{T} y_{it} &= \frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\sum_{t=1}^{T}\alpha_i\gamma_t + \frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\sum_{t=1}^{T}\varepsilon_{it} \\
&= \left(\frac{1}{\sqrt{N}}\sum_{i=1}^{N}\alpha_i\right)\left(\frac{1}{\sqrt{T}}\sum_{t=1}^{T}\gamma_t\right) + \frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\sum_{t=1}^{T}\varepsilon_{it} \\
&\xrightarrow{d} Z_1 Z_2 + Z_3
\end{aligned}
$$

where the Z's denote normally distributed random variables. Product of normally distributed random variables is not normal.

Example 2. *Suppose $y_{it} = \alpha_i + \gamma_t + \alpha_i\gamma_t + \varepsilon_{it}$. Use $N = T$ for exposition. α_i, γ_t, and ε_{it} are iid*

mean zero.

$$\begin{aligned}
\sqrt{N}\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T} y_{it} &= \sqrt{N}\frac{1}{NT}\sum_{i,t}\left(\alpha_i+\gamma_t+\alpha_i\gamma_t+\varepsilon_{it}\right) \\
&= \frac{1}{\sqrt{N}}\sum_{i=1}^{N}\alpha_i + \frac{1}{\sqrt{T}}\sum_{t=1}^{T}\gamma_t + \sqrt{N}\left(\frac{1}{N}\sum_{i=1}^{N}\alpha_i\right)\left(\frac{1}{T}\sum_{t=1}^{T}\gamma_t\right) + \frac{1}{\sqrt{T}}\left(\frac{1}{\sqrt{NT}}\sum_{i,t}\varepsilon_{it}\right) \\
&= \frac{1}{\sqrt{N}}\sum_{i=1}^{N}\alpha_i + \frac{1}{\sqrt{T}}\sum_{t=1}^{T}\gamma_t + \frac{1}{\sqrt{N}}\sum_{i=1}^{N}\alpha_i o_P(1) + o_P(1)O_P(1) \\
&\xrightarrow{d} Z_1 + Z_2
\end{aligned}$$

This pair of examples is informative of how the proof strategy might work. In Example 2, having the first two terms be orthogonal to each other and asymptotically normal while the interaction term is $o_P(1)$ is sufficient to give normality. If the asymptotics are driven by the interaction term as in Example 1, then we cannot guarantee normality. Hence, the appropriate assumption allows asymptotics to be driving by the first two terms.

Assumption 4. *Either* $\lim_{N\to\infty}\frac{\sum_g\left(\frac{N_g^G}{N}\right)^{2+\delta}}{\left(\sum_g(\tilde{\varepsilon}_g^w)^2\left(\frac{N_g^G}{N}\right)^2\right)^{(2+\delta)/2}}=0$ *or* $\lim_{N\to\infty}\frac{\sum_h\left(\frac{N_h^H}{N}\right)^{2+\delta}}{\left(\sum_h(\tilde{\varepsilon}_h^w)^2\left(\frac{N_h^H}{N}\right)^2\right)^{(2+\delta)/2}}=0$*, or both.*

The shortcoming of the assumption is that it makes a statement about ε instead of the primitives of the model. It is possible to state a stronger assumption on the primitives of the model that implies Assumption 4.

Assumption 5. *The following hold:*

(i) As $N\to\infty$, either there are finitely many clusters g such that, for all w, $\frac{1}{N_g^G}\sum_{i\in S_g}Y_i(w)-\bar{Y}(w)=0$, or there are finitely many clusters h such that, for all w, $\frac{1}{N_h^H}\sum_{i\in S_h}Y_i(w)-\bar{Y}(w)=0$.

(ii) $\sum_g\left(\frac{(N_g^G)^2}{\sum_{g'}(N_{g'}^G)^2}\right)^2\to 0$ and $\sum_h\left(\frac{(N_h^H)^2}{\sum_{h'}(N_{h'}^H)^2}\right)^2\to 0$.

Lemma 4. *Under Assumption 2, Assumption 5 implies Assumption 4.*

When Assumption 4 is satisfied, asymptotics will be driven by either of the first two terms because the Lindeberg condition is satisfied, allowing the application of CLT. This assumption jointly makes

a statement about the potential outcomes and the sequence of cluster sizes. As a benchmark, Hansen and Lee (2019) required $\max_g \frac{(N_g^G)^2}{N} \to 0$ for asymptotic normality, as well as an assumption on the limiting expectation that allows the Lindeberg condition to hold. The first part of their assumption is too strong for purpose, in light of Assumption 3. Their latter assumption is essentially captured by Assumption 4 in our context.

Assumption 5(i) rules out having too many $\varepsilon = 0$ objects, which is required in Assumption 4. In contrast, we did not require this assumption in prior consistency proofs. If errors are zero after a while, the first few terms can still converge to zero, since 1/N still decreases. Having a large chain without error is also good for consistency because we are essentially getting repeated observations of the truth.

Assumption 4 failing does not mean that inference is a hopeless task. It merely means that we do not have guarantees of asymptotic normality, and other inference procedures like bootstrapping are required. Without asymptotic normality, it is then not obvious why we want to use the standard clustered variance estimators stated above.

Theorem 3. *Under Assumptions 1 to 4, either* $s_N = 0$ *or* $\frac{\sqrt{\mu(1-\mu)}}{s_N}(\hat{\tau} - \tau) \xrightarrow{d} N(0, \Omega)$.

The proof proceeds by using the Aldous-Hoover representation, and mimicking Example 2 above. To split the interaction term into a product, the spectral representation theorem is employed. The representation is valid due to our binary assignment and bounded outcomes that implies L^2 integrability. Then, it remains to show that the excess terms are indeed $o_P(1)$. The form of Ω is not crucial here because we have already derived the true variance in Lemma 5. The theorem merely shows that the estimator is indeed asymptotically normal.

Observe that Assumption 4 does not rule out $s_N = 0$ because we can have $V_G = 0$ or $V_H = 0$. We require at least one of the Q objects to have positive variance in order to yield a nonzero s_N. It is insufficient to have $var(A_g) > 0$ because $f(.)$ may send it to a flat region. For instance, A_g can be 1 or 2 with equal probability, but $f(A_g, B_h, v) = 1[A_g \geq 0]B_h$, and Q^G will have zero variance. When either of the Q objects has positive variance, asymptotics can be driven by the first two terms of the Aldous-Hoover representation, thereby averting pathological cases of non-normality. This is similar to the normality condition in MacKinnon et al. (2021).

By construction of s_N, there is no need to impose any particular rate of convergence. As noted

in Example 5 of Hansen and Lee (2019), the convergence rate can be as fast as $N^{-1/2}$, and can be slower than $(M_N^G)^{-1/2}$. The slow rate of convergence occurs when the number of clusters is determined by a large number of small clusters, while the convergence rate is determined by a small number of large clusters.

5 Discussion: Practical Implications and Takeaways

The main empirical takeaway from the paper is that, when researchers are interested in the sample ATE, they should cluster only on dimensions that affect assignment. This should inform future empirical practice. For instance, in randomized controlled trials in development economics, as long as assignment is independent over individuals, one should use EHW errors instead of clustering by villages, even if their outcomes are correlated. EHW errors are sufficient to characterize the uncertainty for sample ATE.

Even if the researcher is interested in the population ATE, or is interested in a model-based framework, it is still valuable to report standard errors in a design-based framework, which reports the uncertainty in sample ATE. Making inferences on the population ATE requires further assumptions on how the sample is drawn from the population, whereas the sample ATE is a well-defined object without requiring such assumptions. If the researcher has a significant result with EHW standard errors and an insignificant result with CGM standard errors, it just means that she can say something about the sample, but not the population. With population data, this phenomenon implies she can make valid inferences about the population at hand, but not the superpopulation that the population is obtained from.

Even in light of the design-based framework, I have shown that the OLS estimator is consistent and asymptotically normal. Standard variance estimators like EHW, LZ and CGM also converge to their asymptotic variance. CGM variance is conservative in general, but when we carefully consider how treatment is assigned, we can appropriately choose which dimensions to cluster on.

A Proofs

A.1 Proofs for Section 4.1

Proof of Lemma 1. The proof proceeds by applying Theorem 2 of von Bahr and Esseen (1965), which requires a martingale difference sequence and the existence of $1+\delta$ moments. Mean independence of X_i implies $X_i - \mu_i$ is a martingale difference sequence. To see this, $E[X_i - \mu_i | \sum_{k=1}^{i-1}(X_k - \mu_k)] = E\{E[X_i - \mu_i | X_j - \mu_j, j < i] | \sum_{k=1}^{i-1}(X_k - \mu_k)\} = 0$.

Let $S_n := \sum_{i=1}^{n} a_{ni}(X_i - \mu_i)$. By Markov's inequality, $\forall \varepsilon > 0$,

$$\begin{aligned} P(|S_n| \geq \varepsilon) = P(|S_n|^{1+\delta} \geq \varepsilon^{1+\delta}) &\leq \frac{E|S_n|^{1+\delta}}{\varepsilon^{1+\delta}} \\ &\leq \frac{\sum_i a_{ni}^{1+\delta} E|X_i - \mu_i|^{1+\delta}}{\varepsilon^{1+\delta}} \\ &\leq \frac{\sum_i a_{ni}^{1+\delta} K}{\varepsilon^{1+\delta}} \xrightarrow{p} 0 \end{aligned}$$

The second inequality comes from the von Bahr and Esseen (1965) result. The final inequality uses the fact that we have finite and bounded $1+\delta$ moments, so $\exists K < \infty$ such that $\max_i E|X_i - \mu_i|^{1+\delta} < K$. Finally, the assumption that $\sum_{i=1}^{n} a_{ni}^{1+\delta} \to 0$ yields convergence. □

Proof of Lemma 2. Using the notation of Equation 3, due to independence in Assumption 1, and the fact that transformations of independent variables are also independent, Q_g^G and Q_h^H are mutually independent. Under the Aldous-Hoover representation, I will define the following objects with slight abuse of notation.

$$\begin{aligned} W_i &= \mu + a_{G_i} + b_{H_i} + c_{G_i H_i} + u_i \\ a_g &= E[W_i | A_g] - \mu = Q_g^G - \mu \\ b_h &= E[W_i | B_h] - \mu = Q_h^H - \mu \\ c_{gh} &= E[W_i | A_g, B_h] - E[W_i | A_g] - E[W_i | B_h] + \mu \\ &= Q_{gh}^{GH} - Q_g^G - Q_h^H + \mu \\ u_i &= W_i - E[W_i | A_g, B_h] \end{aligned}$$

These definitions apply only to this proof. In the rest of the paper, the a, b, c objects are defined for the Aldous-Hoover representation of $T_i\varepsilon_i$.

Using the representation, consider the terms individually, as $\bar{W} := \frac{1}{N}\sum_i W_i = \mu + \bar{a} + \bar{b} + \bar{c} + \bar{u}$.

$$\bar{a} = \frac{1}{N}\sum_{i=1}^{N} a_{G_i} = \sum_g \frac{N_g^G}{N}(Q_g^G - \mu)$$

Assumption 2(ii) ensures that the weights $\frac{N_g^G}{N}$ satisfy the condition for WLLN so Lemma 1 can be applied. Clearly, W_i is bounded because it is binary. Since $E[Q_g^G] = \mu$, as the number of clusters G get large by Assumption 2(i), we obtain $\bar{a} = o_P(1)$. The argument for $\bar{b} = o_P(1)$ is completely analogous.

Next, take care of the product term. Applying Assumption 1 as before, $Q_h^H - \mu$ is independent over h. It turns out that LLN can also be applied on the sum of $\sum_g \frac{N_{gh}}{N_h^H}(Q_{gh}^{GH} - Q_g^G)$ over h, though the derivation is more involved. For $g \neq g'$ and $h \neq h'$, mean independence of $Q_{gh}^{GH} - Q_g^G$ clearly holds. Next, since $E_h[Q_{gh}^{GH}] = Q_g^G$, observe that:

$$\begin{aligned} E[Q_{gh}^{GH} - Q_g^G | Q_{gh'}^{GH} - Q_g^G] &= E_g\{E_h[Q_{gh}^{GH} - Q_g^G | Q_{gh'}^{GH} - Q_g^G, g]\} \\ &= E_g[0] = 0 \end{aligned}$$

Mean independence is sufficient to apply WLLN. Finite moments is automatic due to the binary assignment and $\sum_g \frac{N_{gh}}{N_h^H} = 1$, so $E[\sum_g \frac{N_{gh}}{N_h^H}|Q_{gh}^{GH} - Q_g^G|] < \infty$ by Cauchy-Schwarz inequality. Hence, $\sum_h \frac{N_h^H}{N}\sum_g \frac{N_{gh}}{N_h^H}(Q_{gh}^{GH} - Q_g^G) = o_P(1)$. Then,

$$\begin{aligned} \bar{c} &:= \frac{1}{N}\sum_{i=1}^{N} c_{G_iH_i} = \frac{1}{N}\sum_{i=1}^{N} Q_{G_iH_i}^{GH} - Q_{G_i}^G - Q_{H_i}^H + \mu \\ &= \sum_h \frac{N_h^H}{N}\sum_g \frac{N_{gh}}{N_h^H}(Q_{gh}^{GH} - Q_g^G) - \sum_h \frac{N_h^H}{N}(Q_h^H - \mu) = o_P(1) \end{aligned}$$

A similar argument can be applied to the u_i term by first showing that the terms are mean independent, then applying the WLLN. It is sufficient to prove mean independence of u_i, u_j when i, j

share both the g and h clusters. If they are in different clusters, an analogous argument can be applied.

$$
\begin{aligned}
u_i &= f(A_{G_i}, B_{H_i}, v_i) - E_v[f(A_{G_i}, B_{H_i}, v_i)] \\
E[u_i|u_j] &= E_{g,h}\{E[u_i|u_j, g, h]\} \\
&= E_{g,h}[f(A_g, B_h, v_i) - E_v[f(A_g, B_h, v_i)]|f(A_g, B_h, v_j) - E_v[f(A_g, B_h, v_j), g, h]] \\
&= E_{g,h}[0] = 0
\end{aligned}
$$

Hence, $\bar{u} = o_P(1)$. To complete the argument, using continuous mapping theorem (CMT), $\bar{W} = \mu + o_P(1)$. □

Proof of Lemma 3. Apply $\frac{1}{N}\sum_{i=1}^N W_i = \mu + o_P(1)$ in the denominator.[4]

Then,

$$
\begin{aligned}
\hat{\tau} - \tau &= \left(\frac{\frac{1}{N}\sum_{i=1}^N W_i Y_i(1)}{\frac{1}{N}\sum_{i=1}^N W_i} - \bar{Y}(1)\right) - \left(\frac{\frac{1}{N}\sum_{i=1}^N (1-W_i) Y_i(0)}{1 - \frac{1}{N}\sum_{i=1}^N W_i} - \bar{Y}(0)\right) \\
&= \left(\frac{\frac{1}{N}\sum_{i=1}^N W_i(\varepsilon_i(1) + \bar{Y}(1))}{\frac{1}{N}\sum_{i=1}^N W_i} - \bar{Y}(1)\right) - \left(\frac{\frac{1}{N}\sum_{i=1}^N (1-W_i)(\varepsilon_i(0) + \bar{Y}(0))}{1 - \frac{1}{N}\sum_{i=1}^N W_i} - \bar{Y}(0)\right) \\
&= \left(\frac{\frac{1}{N}\sum_{i=1}^N W_i\varepsilon_i(1) + \mu\bar{Y}(1)}{\mu} - \bar{Y}(1)\right) - \left(\frac{\frac{1}{N}\sum_{i=1}^N (1-W_i)\varepsilon_i(0) + (1-\mu)\bar{Y}(0)}{1-\mu} - \bar{Y}(0)\right) + o_P(1) \\
&= \frac{\frac{1}{N}\sum_{i=1}^N W_i\varepsilon_i(1)}{\mu} - \frac{\frac{1}{N}\sum_{i=1}^N (1-W_i)\varepsilon_i(0)}{1-\mu} + o_P(1) \\
&= \frac{\frac{1}{N}\left[\sum_{i=1}^N W_i\varepsilon_i(1) - \mu\sum_{i=1}^N (W_i\varepsilon_i(1) + (1-W_i)\varepsilon_i(0))\right]}{\mu(1-\mu)} + o_P(1)
\end{aligned}
$$

[4]Note that, for nonzero constants a, b,

$$
\frac{a}{b + o_P(1)} = \frac{\frac{a}{b}(b + o_P(1) - o_P(1))}{b + o_P(1)} = \frac{a}{b} - \frac{\frac{a}{b}o_P(1)}{b + o_P(1)} = \frac{a}{b} + o_P(1)
$$

Using the fact that $\sum_i W_i\varepsilon_i = \sum_i W_i\varepsilon_i(1)$ and $\sum_i \varepsilon_i = \sum_i W_i\varepsilon_i(1) + (1-W_i)\varepsilon_i(0)$,

$$\begin{aligned}\hat{\tau} - \tau &= \frac{\frac{1}{N}\left(\sum_{i=1}^N W_i\varepsilon_i - \mu\sum_{i=1}^N \varepsilon_i\right)}{\mu(1-\mu)} + o_P(1) \\ &= \frac{\frac{1}{N}\sum_{i=1}^N (W_i-\mu)\varepsilon_i}{\mu(1-\mu)} + o_P(1)\end{aligned}$$

To transform W into a mean zero and variance one object, use $T_i := \frac{W_i-\mu}{\sqrt{\mu(1-\mu)}}$. Then, with $\eta_i := T_i\varepsilon_i$, obtain:

$$\begin{aligned}\hat{\tau} - \tau &= \frac{1}{N\sqrt{\mu(1-\mu)}}\sum_{i=1}^N T_i\varepsilon_i + o_P(1) \\ &= \frac{1}{N\sqrt{\mu(1-\mu)}}\sum_{i=1}^N \eta_i + o_P(1)\end{aligned}$$

□

Proof of Theorem 1. The error terms can be written as nonstochastic $\varepsilon_i(w)$ objects in the potential outcomes framework. Let $\tilde{\eta}_i := \eta_i - E[\eta_i]$.

$$\begin{aligned}E[\varepsilon_i] &= \mu\varepsilon_i(1) + (1-\mu)\varepsilon_i(0) \\ E[\varepsilon_i^2] &= \mu\varepsilon_i(1)^2 + (1-\mu)\varepsilon_i(0)^2 \\ E[\eta_i] &= \mu\sqrt{\frac{1-\mu}{\mu}}\varepsilon_i(1) - (1-\mu)\sqrt{\frac{\mu}{1-\mu}}\varepsilon_i(0) \\ &= \sqrt{\mu(1-\mu)}(\varepsilon_i(1) - \varepsilon_i(0)) \\ E[\eta_i^2] &= \mu\cdot\frac{1-\mu}{\mu}\varepsilon_i(1)^2 + (1-\mu)\cdot\frac{\mu}{1-\mu}\varepsilon_i(0)^2 \\ &= (1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2 \\ E[\tilde{\eta}_i^2] &= (1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2 - \mu(1-\mu)(\varepsilon_i(1) - \varepsilon_i(0))^2 \\ &= (1-\mu)^2\varepsilon_i(1)^2 + \mu^2\varepsilon_i(0)^2 + 2\mu(1-\mu)\varepsilon_i(1)\varepsilon_i(0) \\ &= ((1-\mu)\varepsilon_i(1) + \mu\varepsilon_i(0))^2\end{aligned}$$

Due to the approximation in Lemma 3, it is sufficient to show:

$$\frac{1}{N}\sum_{i=1}^{N}\eta_i \xrightarrow{p} 0$$

Apply the Aldous-Hoover representation and use the $Q_g^G := E_{h,v}[f(A_g, B_h, v)]$ notation as before.

$$\begin{aligned}
\tilde{\eta}_i &= a_{G_i} + b_{H_i} + c_{G_iH_i} + e_i \\
a_{G_i} &= E[\eta_i|A_{G_i}] - \sqrt{\mu(1-\mu)}(\varepsilon_i(1) - \varepsilon_i(0)) \\
&= Q_{G_i}^G\sqrt{\frac{1-\mu}{\mu}}\varepsilon_i(1) - (1-Q_{G_i}^G)\sqrt{\frac{\mu}{1-\mu}}\varepsilon_i(0) - \sqrt{\mu(1-\mu)}(\varepsilon_i(1) - \varepsilon_i(0))
\end{aligned}$$

Then, a_{G_i} has mean zero. Further, we want to aggregate the a_{G_i} within each cluster so that we get independence over clusters. This stems from the observation that Q_g^G is independent over g, regardless of the distribution of H.

Let S_g denote the set of individuals that are in cluster g. Define cluster specific residuals as $\tilde{\varepsilon}_g(w) := \frac{1}{N_g^G}\sum_{i\in S_g}\varepsilon_i(w)$. Then, a_g can be written as:

$$\begin{aligned}
a_g &= \frac{1}{N_g^G}\sum_{i\in S_g} a_{G_i} \\
&= Q_g^G\sqrt{\frac{1-\mu}{\mu}}\tilde{\varepsilon}_g(1) - (1-Q_g^G)\sqrt{\frac{\mu}{1-\mu}}\tilde{\varepsilon}_g(0) - \sqrt{\mu(1-\mu)}(\tilde{\varepsilon}_g(1) - \tilde{\varepsilon}_g(0)) \\
&= Q_g^G\left(\sqrt{\frac{1-\mu}{\mu}}\tilde{\varepsilon}_g(1) + \sqrt{\frac{\mu}{1-\mu}}\tilde{\varepsilon}_g(0)\right) - \sqrt{\frac{\mu}{1-\mu}}\tilde{\varepsilon}_g(0) - \sqrt{\mu(1-\mu)}(\tilde{\varepsilon}_g(1) - \tilde{\varepsilon}_g(0)) \\
&= Q_g^G\left(\sqrt{\frac{1-\mu}{\mu}}\tilde{\varepsilon}_g(1) + \sqrt{\frac{\mu}{1-\mu}}\tilde{\varepsilon}_g(0)\right) - \mu\left[\sqrt{\frac{1-\mu}{\mu}}\tilde{\varepsilon}_g(1) - \tilde{\varepsilon}_g(0)(-\mu/(1-\mu))\sqrt{\frac{1-\mu}{\mu}}\right] \\
&= (Q_g^G - \mu)\left(\sqrt{\frac{1-\mu}{\mu}}\tilde{\varepsilon}_g(1) + \sqrt{\frac{\mu}{1-\mu}}\tilde{\varepsilon}_g(0)\right)
\end{aligned}$$

Check the condition for WLLN. Use notation $\tilde{\varepsilon}_g^w := \left(\sqrt{\frac{1-\mu}{\mu}}\tilde{\varepsilon}_g(1) + \sqrt{\frac{\mu}{1-\mu}}\tilde{\varepsilon}_g(0)\right)$. Assumption 2(iii) implies that $E[|(Q_g^G - \mu)\tilde{\varepsilon}_g^w|^{1+\delta}] < \infty$, which is immediate from bounded outcomes and binary assignment.

$$\bar{a} = \frac{1}{N}\sum_{i=1}^{N} a_{G_i} = \sum_g \frac{N_g^G}{N} a_g = \sum_g \frac{N_g^G}{N}(Q_g^G - \mu)\tilde{\varepsilon}_g^w$$

Since weights satisfy $\sum_g \left(\frac{N_g^G}{N}\right)^{1+\delta} \to 0$ by Assumption 2(ii), use Lemma 1 as the number of clusters get large by Assumption 2(i). Hence, $\sum_g (N_g^G/N)a_g = o_P(1)$. The same representation applies to b_{H_i} and hence b_h. Then, $\bar{a} = o_P(1)$ and $\bar{b} = o_P(1)$.

The cross term is:

$$\begin{aligned} c_{G_iH_i} &= E[\eta_i|A_{G_i}, B_{H_i}] - E[\eta_i|A_{G_i}] - E[\eta_i|B_{H_i}] + \sqrt{\mu(1-\mu)}(\varepsilon_i(1) - \varepsilon_i(0)) \\ &= Q_{G_iH_i}^{GH}\sqrt{\frac{1-\mu}{\mu}}\varepsilon_i(1) - (1 - Q_{G_iH_i}^{GH})\sqrt{\frac{\mu}{1-\mu}}\varepsilon_i(0) - E[\eta_i|A_{G_i}] - E[\eta_i|B_{H_i}] \\ &\quad + \sqrt{\mu(1-\mu)}(\varepsilon_i(1) - \varepsilon_i(0)) \end{aligned}$$

When aggregating, we deal with the middle terms $E[\eta_i|A_{G_i}], E[\eta_i|B_{H_i}]$ slightly differently from before due to the presence of both g and h indices. Namely,

$$\begin{aligned} \frac{1}{N_{gh}}\sum_{i\in S_{gh}} a_{G_i} &= Q_g^G\sqrt{\frac{1-\mu}{\mu}}\tilde{\varepsilon}_{gh}(1) - (1 - Q_g^G)\sqrt{\frac{\mu}{1-\mu}}\tilde{\varepsilon}_{gh}(0) - \sqrt{\mu(1-\mu)}(\tilde{\varepsilon}_{gh}^w(1) - \tilde{\varepsilon}_{gh}(0)) \\ &= (Q_g^G - \mu)\left(\sqrt{\frac{1-\mu}{\mu}}\tilde{\varepsilon}_{gh}(1) + \sqrt{\frac{\mu}{1-\mu}}\tilde{\varepsilon}_{gh}(0)\right) \end{aligned}$$

The error terms are now indexed by both g and h instead of just g. Since $E[\eta_i|A_{G_i}] = a_{G_i} + \sqrt{\mu(1-\mu)}(\varepsilon_i(1) - \varepsilon_i(0))$, some algebra using an analogous method yields:

$$c_{gh} = \frac{1}{N_{gh}}\sum_{i\in S_{gh}} c_{G_iH_i} = (Q_{gh}^{GH} - Q_g^G - Q_h^H + \mu)\left(\sqrt{\frac{1-\mu}{\mu}}\tilde{\varepsilon}_{gh}(1) + \sqrt{\frac{\mu}{1-\mu}}\tilde{\varepsilon}_{gh}(0)\right)$$

Then, using a similar method as Lemma 2 for the c term yields:

$$\begin{aligned} \bar{c} &:= \frac{1}{N}\sum_{i=1}^{N} c_{G_iH_i} = \frac{1}{N}\sum_{i=1}^{N}(Q_{G_iH_i}^{GH} - Q_{G_i}^G - Q_{H_i}^H + \mu)\left(\sqrt{\frac{1-\mu}{\mu}}\varepsilon_i(1) + \sqrt{\frac{\mu}{1-\mu}}\varepsilon_i(0)\right) \\ &= \sum_h \frac{N_h^H}{N}\sum_g \frac{N_{gh}}{N_h^H}(Q_{gh}^{GH} - Q_g^G)\tilde{\varepsilon}_{gh}^w - \sum_h \frac{N_h^H}{N}(Q_h^H - \mu)\tilde{\varepsilon}_{gh}^w = o_P(1) \end{aligned}$$

Asymptotics in the last line relies on Assumption 2(iii) which implies bounded moments. In the final term of the representation, e_i is treated in the same way as u_i in Lemma 2. Consequently, $\frac{1}{N}\sum_{i=1}^{N}\tilde{\eta}_i = o_P(1)$.

To complete the argument, since $\sum_{i=1}^{N}\varepsilon_i(w) = 0$, observe that:

$$\begin{aligned}\frac{1}{N}\sum_{i=1}^{N}\eta_i &= \frac{1}{N}\sum_{i=1}^{N}\eta_i + \frac{1}{N}\sum_{i=1}^{N}\sqrt{\mu(1-\mu)}(\varepsilon_i(1)-\varepsilon_i(0)) \\ &= \frac{1}{N}\sum_{i=1}^{N}(\eta_i - E[\eta_i]) = \frac{1}{N}\sum_{i=1}^{N}\tilde{\eta}_i \xrightarrow{p} 0\end{aligned}$$

□

A.2 Proofs for Section 4.2

Lemma 5. *Under Assumption 1, variance of the approximated object is given by:*

$$\begin{aligned}\mathbb{V}_\eta = \frac{1}{N^2\mu(1-\mu)}\{&\sum_g (N_g^G)^2 V_G(\tilde{\varepsilon}_g^w)^2 + \sum_h (N_h^H)^2 V_H(\tilde{\varepsilon}_h^w)^2 \\ &+ \sum_{g,h}(N_{gh}^2)[V_{GH}(\tilde{\varepsilon}_{gh}^w)^2 - V_G(\tilde{\varepsilon}_g^w)^2 - V_H(\tilde{\varepsilon}_h^w)^2] + \sum_i [(1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2 - \nu_i^2 - V_{GH}(\varepsilon_i^w)^2]\}\end{aligned}$$

Proof of Lemma 5. Normalized variance of the approximated object is

$$
\begin{aligned}
\mathbb{V}_\eta &= \frac{1}{N^2\mu(1-\mu)} E\left[\left(\sum_{i=1}^{N}\tilde{\eta}_i\right)^2\right] = \frac{1}{N^2\mu(1-\mu)} E\left[\left(\sum_{i=1}^{N}\tilde{\eta}_i\right)\left(\sum_{j=1}^{N}\tilde{\eta}_j\right)\right] \\
&= \frac{1}{N^2\mu(1-\mu)}\{\sum_{i,j} E[\tilde{\eta}_i\tilde{\eta}_j|G_i=G_j, H_i=H_j, i\neq j]1[G_i=G_j, H_i=H_j, i\neq j] + \sum_i E[\tilde{\eta}_i^2] \\
&\quad + \sum_{i,j} E[\tilde{\eta}_i\tilde{\eta}_j|G_i=G_j, H_i\neq H_j, i\neq j]1[G_i=G_j, H_i\neq H_j, i\neq j] \\
&\quad + \sum_{i,j} E[\tilde{\eta}_i\tilde{\eta}_j|G_i\neq G_j, H_i=H_j, i\neq j]1[G_i\neq G_j, H_i=H_j, i\neq j] \\
&\quad + \sum_{i,j} E[\tilde{\eta}_i\tilde{\eta}_j|G_i\neq G_j, H_i\neq H_j, i\neq j]1[G_i\neq G_j, H_i\neq H_j, i\neq j]\} \\
&= \frac{1}{N^2\mu(1-\mu)}\{\sum_{g,h}(N_{gh}^2 - N)E[\tilde{\eta}_i\tilde{\eta}_j|G_i=G_j=g, H_i=H_j=h, i\neq j] + \sum_i E[\tilde{\eta}_i^2] \\
&\quad + \sum_g \left((N_g^G)^2 - \sum_h N_{gh}^2\right) E[\tilde{\eta}_i\tilde{\eta}_j|G_i=G_j=g, H_i\neq H_j, i\neq j] \\
&\quad + \sum_h \left((N_h^H)^2 - \sum_g N_{gh}^2\right) E[\tilde{\eta}_i\tilde{\eta}_j|G_i\neq G_j, H_i=H_j=h, i\neq j]\}
\end{aligned}
$$

Observe that the last term drops because $E[\tilde{\eta}_i\tilde{\eta}_j|G_i\neq G_j, H_i\neq H_j, i\neq j] = E[\tilde{\eta}_i]^2 = 0$.

$$
E[\tilde{\eta}_i\tilde{\eta}_j|g,h] = E[(\eta_i-\nu_i)(\eta_j-\nu_j)|g,h] = E[\eta_i\eta_j|g,h] - \nu_i\nu_j \tag{7}
$$

Using Equation 7 and the respective expressions for conditional expectations from the derivation of Lemma 6,

$$
\begin{aligned}
N^2\mu(1-\mu)\mathbb{V}_\eta &= \sum_g (N_g^G)^2 V_G(\tilde{\varepsilon}_g^w)^2 + \sum_h (N_h^H)^2 V_H(\tilde{\varepsilon}_h^w)^2 \\
&\quad + \sum_{g,h}(N_{gh}^2)[V_{GH}(\tilde{\varepsilon}_{gh}^w)^2 - V_G(\tilde{\varepsilon}_g^w)^2 - V_H(\tilde{\varepsilon}_h^w)^2] + \sum_i[(1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2 - \nu_i^2 - V_{GH}(\varepsilon_i^w)^2]
\end{aligned}
$$

□

Lemma 6. *Under Assumption 1 and 2, asymptotic variance is given by:*

$$\mathbb{V}_{EHW} = \frac{1}{N^2\mu(1-\mu)} \sum_{i=1}^{N} (1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2$$

$$\begin{aligned}
\mathbb{V}_{LZG} = \frac{1}{N^2\mu(1-\mu)} \{ & \sum_g (N_g^G)^2(\tilde{\nu}_g^2 + V_G(\tilde{\varepsilon}_g^w)^2) + \sum_{g,h} (N_{gh}^2)[(\tilde{\nu}_{gh}^2 + V_{GH}(\tilde{\varepsilon}_{gh}^w)^2) - (\tilde{\nu}_g^2 + V_G(\tilde{\varepsilon}_g^w)^2)] \\
& + \sum_i [(1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2 - \nu_i^2 - V_{GH}(\varepsilon_i^w)^2]\}
\end{aligned}$$

$$\begin{aligned}
\mathbb{V}_{CGM} = \frac{1}{N^2\mu(1-\mu)} \{ & \sum_g (N_g^G)^2(\tilde{\nu}_g^2 + V_G(\tilde{\varepsilon}_g^w)^2) + \sum_h (N_h^H)^2(\tilde{\nu}_h^2 + V_H(\tilde{\varepsilon}_h^w)^2) \\
& + \sum_{g,h} (N_{gh}^2)[(\tilde{\nu}_{gh}^2 + V_{GH}(\tilde{\varepsilon}_{gh}^w)^2) - (\tilde{\nu}_g^2 + V_G(\tilde{\varepsilon}_g^w)^2) - (\tilde{\nu}_h^2 + V_H(\tilde{\varepsilon}_h^w)^2)] \\
& + \sum_i [(1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2 - \nu_i^2 - V_{GH}(\varepsilon_i^w)^2]\}
\end{aligned}$$

Proof of Lemma 6. The EHW variance is straightforward:

$$\mathbb{V}_{EHW} = \frac{1}{N^2\mu(1-\mu)} \sum_{i=1}^{N} E[\eta_i^2] = \frac{1}{N^2\mu(1-\mu)} \sum_{i=1}^{N}\sum_{i=1}^{N} (1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2$$

The asymptotic variance of LZ Estimator can be written as:

$$\begin{aligned}
\mathbb{V}_{LZG} &= \frac{1}{N^2\mu(1-\mu)} \sum_g \sum_{i,j} 1[G_i = G_j = g] E[\eta_i\eta_j | G_i = G_j] \\
&= \frac{1}{N^2\mu(1-\mu)} \sum_{g,h} \sum_{i,j} 1[G_i = G_j = g, H_i \neq H_j, i \neq j] E[\eta_i\eta_j | G_i = G_j = g, H_i \neq H_j, i \neq j] \\
&\quad + \frac{1}{N^2\mu(1-\mu)} \sum_{g,h} \sum_{i,j} 1[G_i = G_j = g, H_i = H_j = h, i \neq j] E[\eta_i\eta_j | G_i = G_j = g, H_i = H_j = h, i \neq j] \\
&\quad + \frac{1}{N^2\mu(1-\mu)} \sum_i E[\eta_i^2]
\end{aligned}$$

For accounting, note that there are $\sum_g \left((N_g^G)^2 - \sum_h N_{gh}^2 \right)$ of the $E[\eta_i\eta_j | G_i = G_j = g, H_i \neq H_j, i \neq j]$ objects and $\sum_{g,h}(N_{gh}^2 - N)$ of the $E[\eta_i\eta_j | G_i = G_j = g, H_i = H_j = h, i \neq j]$ objects. The strategy for the rest of this proof is to work out the $E[\eta_i\eta_j | G_i = G_j = g, H_i \neq H_j, i \neq j]$ and $E[\eta_i\eta_j | G_i = G_j = g, H_i = H_j = h, i \neq j]$ objects individually, then combine them to get the required expression.

Recall the Aldous-Hoover representation for η_i with:

$$\begin{aligned}
\eta_i &= \nu_i + a_{G_i} + b_{H_i} + c_{G_iH_i} + e_i \\
a_{G_i} &= (Q^G_{G_i} - \mu)\left(\sqrt{\frac{1-\mu}{\mu}}\varepsilon_i(1) + \sqrt{\frac{\mu}{1-\mu}}\varepsilon_i(0)\right) \\
b_{H_i} &:= (Q^H_{H_i} - \mu)\left(\sqrt{\frac{1-\mu}{\mu}}\varepsilon_i(1) + \sqrt{\frac{\mu}{1-\mu}}\varepsilon_i(0)\right) \\
c_{G_iH_i} &:= (Q^{GH}_{G_iH_i} - Q^G_{G_i} - Q^H_{H_i} + \mu)\left(\sqrt{\frac{1-\mu}{\mu}}\varepsilon_i(1) + \sqrt{\frac{\mu}{1-\mu}}\varepsilon_i(0)\right)
\end{aligned}$$

Work out the cross products and their conditional expectations.

$$\begin{aligned}
\eta_i\eta_j &= (\nu_i + a_{G_i} + b_{H_i} + c_{G_iH_i} + e_i)(\nu_j + a_{G_j} + b_{H_j} + c_{G_jH_j} + e_j) \\
E[\eta_i\eta_j | G_i \neq G_j, H_i \neq H_j, i \neq j] &= \left(\sqrt{\mu(1-\mu)}(\varepsilon_i(1) - \varepsilon_i(0))\right)^2 = \mu(1-\mu)(\varepsilon_i(1) - \varepsilon_i(0))^2 \\
E[\eta_i\eta_j | G_i = G_j, H_i = H_j, i \neq j] & \\
&= E[(\nu_i + a_{G_i} + b_{H_i} + c_{G_iH_i} + e_i)(\nu_j + a_{G_j} + b_{H_j} + c_{G_jH_j} + e_j) | G_i = G_j, H_i = H_j, i \neq j] \\
&= \nu_i\nu_j + E[(a_{G_i} + b_{H_i} + c_{G_iH_i})(a_{G_j} + b_{H_j} + c_{G_jH_j}) | G_i = G_j, H_i = H_j, i \neq j]
\end{aligned}$$

The final equality follows from $E[\nu_i(a_{G_j} + b_{H_j} + c_{G_jH_j}) | G_i = G_j, H_i = H_j, i \neq j] = 0$ since ν_i is non stochastic.

Define covariances in the following way:

$$\begin{aligned}
V_{CG} &:= E[(Q^{GH}_{gh} - \mu)(Q^G_g - \mu)] \\
V_{CH} &:= E[(Q^{GH}_{gh} - \mu)(Q^H_h - \mu)]
\end{aligned}$$

$$\begin{aligned}
E[a_{G_i}a_{G_j} | G_i = G_j] &= E[(Q^G_g - \mu)^2 \varepsilon^w_i \varepsilon^w_j] = V_G \varepsilon^w_i \varepsilon^w_j \\
E[b_{H_i}b_{H_j} | H_i = H_j] &= V_H \varepsilon^w_i \varepsilon^w_j \\
E[a_{G_i}b_{H_j} | G_i = G_j, H_i = H_j] &= 0
\end{aligned}$$

The last equality occurs due to mutual independence of Q^G_g and Q^H_h.

$$
\begin{aligned}
E[c_{G_iH_i}c_{G_jH_j}|G_i = G_j, H_i = H_j] &= E[(Q_{gh}^{GH} - Q_g^G - Q_h^H + \mu)^2]\varepsilon_i^w\varepsilon_j^w \\
&= E[(Q_{gh}^{GH} - Q_g^G)^2 + (Q_h^H - \mu)^2 - 2(Q_{gh}^{GH} - Q_g^G)(Q_h^H - \mu)]\varepsilon_i^w\varepsilon_j^w \\
&= E[(Q_{gh}^{GH} - \mu)^2 + (Q_g^G - \mu)^2 - 2(Q_{gh}^{GH} - Q_g^G)(Q_g^G - \mu)] + (V_H - 2V_{CH})\varepsilon_i^w\varepsilon_j^w \\
&= (V_{GH} + V_G + V_H - 2V_{CG} - 2V_{CH})\varepsilon_i^w\varepsilon_j^w
\end{aligned}
$$

$$
\begin{aligned}
E[a_{G_i}c_{G_iH_i}|G_i = G_j, H_i = H_j] &= E[(Q_g - \mu)(Q_{gh}^{GH} - Q_g^G - Q_h^H + \mu)]\varepsilon_i^w\varepsilon_j^w \\
&= E[(Q_g^G - \mu)(Q_{gh}^{GH} - Q_g^G)]\varepsilon_i^w\varepsilon_j^w + E[(Q_g^G - \mu)(Q_h^H - \mu)]\varepsilon_i^w\varepsilon_j^w \\
&= E[(Q_g^G - \mu)(Q_{gh}^{GH} - \mu)]\varepsilon_i^w\varepsilon_j^w - E[(Q_g^G - \mu)^2]\varepsilon_i^w\varepsilon_j^w \\
&= (V_{CG} - V_G)\varepsilon_i^w\varepsilon_j^w
\end{aligned}
$$

Analogously, $E[a_{G_i}c_{G_iH_i}|G_i = G_j, H_i = H_j] = (V_{CH} - V_H)\varepsilon_i^w\varepsilon_j^w$. Substitute these objects into the original $E[\eta_i\eta_j|G_i = G_j, H_i = H_j, i \neq j]$ expression. This yields:

$$
\begin{aligned}
&E[\eta_i\eta_j|G_i = G_j, H_i = H_j, i \neq j] \\
&\quad = \nu_i\nu_j + V_G\varepsilon_i^w\varepsilon_j^w + V_H\varepsilon_i^w\varepsilon_j^w + (V_{GH} + V_G + V_H - 2V_{CG} - 2V_{CH})\varepsilon_i^w\varepsilon_j^w \\
&\qquad + 2(V_{CG} - V_G)\varepsilon_i^w\varepsilon_j^w + 2(V_{CH} - V_H)\varepsilon_i^w\varepsilon_j^w \\
&\quad = \nu_i\nu_j + V_{GH}\varepsilon_i^w\varepsilon_j^w
\end{aligned}
$$

Consider the cross product when they share exactly one cluster. The expression simplifies using an argument identical to before.

$$
\begin{aligned}
&E[\eta_i\eta_j|G_i = G_j, H_i \neq H_j, i \neq j] \\
&\quad = \nu_i\nu_j + E[(a_{G_i} + b_{H_i} + c_{G_iH_i})(a_{G_j} + b_{H_j} + c_{G_jH_j})|G_i = G_j, H_i \neq H_j, i \neq j]
\end{aligned}
$$

Observe that:

$$E[b_{H_i} b_{H_j} | H_i \neq H_j] = E[b_h]^2 = 0$$
$$E[a_g c_{gH_j} | H_i \neq H_j] = E[a_g c_{gH_j}] = (V_{CG} - V_G)\varepsilon_i^w \varepsilon_j^w$$

When $H_i \neq H_j$, $B_{H_i} \perp B_{H_j}$, so

$$\begin{aligned} E[(Q_{gH_i}^{GH} - \mu)(Q_{H_j}^{H} - \mu)] &= E_{gh}\left[(E_v f(A_g, B_{H_j}, v) - \mu)(E_{g,v} f(A_g, B_{H_i}, v) - \mu)\right] \\ &= E_{gh}[E_v f(A_g, B_h, v) - \mu] E_h[E_{g,v} f(A_g, B_h, v) - \mu] = 0 \end{aligned}$$

This implies:

$$E[b_{H_i} c_{gH_j} | H_i \neq H_j] = 0$$

Next,

$$\begin{aligned} &E[c_{G_i H_i} c_{G_j H_j} | G_i = G_j, H_i \neq H_j] \frac{1}{\varepsilon_i^w \varepsilon_j^w} \\ &\quad = E[(Q_{gH_i}^{GH} - Q_g^G - Q_{H_i}^H + \mu)(Q_{gH_j}^{GH} - Q_g^G - Q_{H_j}^H + \mu) | G_i = G_j = g, H_i \neq H_j] \\ &\quad = E[(Q_{gH_i}^{GH} - Q_g^G)(Q_{gH_j}^{GH} - Q_g^G) | H_i \neq H_j] + E[(Q_{H_i}^H - \mu)(Q_{H_j}^H - \mu) | H_i \neq H_j] \\ &\qquad - 2E[(Q_{gH_i}^{GH} - Q_g^G)(Q_{H_j} - \mu) | H_i \neq H_j] \end{aligned}$$

Evaluate the first term:

$$
\begin{aligned}
&E[(Q^{GH}_{gH_i} - Q^G_g)(Q^{GH}_{gH_j} - Q^G_g)|H_i \neq H_j] \\
&\qquad = E[(Q^{GH}_{gH_i} - \mu)(Q^{GH}_{gH_j} - \mu)|H_i \neq H_j] - 2E[(Q^{GH}_{gH_i} - \mu)(Q^G_g - \mu)] + E[(Q^G_g - \mu)^2] \\
&\qquad = E[(Q^{GH}_{gH_i} - \mu)(Q^{GH}_{gH_j} - \mu)|H_i \neq H_j] - 2V_{CG} + V_G \\
&\qquad = E_g\{E_h[(E_v f(A_g, B_{H_i}, v) - \mu)(E_v f(A_g, B_{H_j}, v) - \mu)]\} - 2V_{CG} + V_G \\
&\qquad = E_g\{(E_{v,h} f(A_g, B_{H_i}, v) - \mu)(E_{v,h} f(A_g, B_{H_j}, v) - \mu)\} - 2V_{CG} + V_G \\
&\qquad = E_g\{(Q^G_g - \mu)^2\} - 2V_{CG} + V_G \\
&\qquad = 2V_G - 2V_{CG}
\end{aligned}
$$

Observe that

$$
\begin{aligned}
E[(Q^{GH}_{gH_i} - Q^G_g)(Q_{H_j} - \mu)|H_i \neq H_j] &= E[(Q^{GH}_{gH_i} - \mu)(Q^H_{H_j} - \mu) - (Q^G_g - \mu)(Q^H_{H_j} - \mu)|H_i \neq H_j] \\
&= E[(Q^{GH}_{gH_i} - \mu)(Q^H_{H_j} - \mu)|H_i \neq H_j] = 0
\end{aligned}
$$

Hence,

$$
E[c_{G_i H_i} c_{G_j H_j}|G_i = G_j, H_i \neq H_j] = (2V_G - 2V_{CG})\varepsilon^w_i \varepsilon^w_j
$$

Combining these results,

$$
\begin{aligned}
&E[\eta_i \eta_j|G_i = G_j, H_i \neq H_j, i \neq j] \\
&\qquad = \nu_i \nu_j + V_G \varepsilon^w_i \varepsilon^w_j + (2V_G - 2V_{CG})\varepsilon^w_i \varepsilon^w_j + 2(V_{CG} - V_G)\varepsilon^w_i \varepsilon^w_j \\
&\qquad = \nu_i \nu_j + V_G \varepsilon^w_i \varepsilon^w_j
\end{aligned}
$$

Observe:

$$
\frac{1}{(N^G_g)^2} \sum_{i,j \in S_g} \nu_i \nu_j = \left(\frac{1}{N^G_g} \sum_{i \in S_g} \nu_i \right) \left(\frac{1}{N^G_g} \sum_{j \in S_g} \nu_j \right) = \tilde{\nu}^2_g
$$

Recall $E[\eta_i^2] = (1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2$. Then, put these results back into the expression for the asymptotic variance:

$$
\begin{aligned}
(N^2\mu(1-\mu))\mathbb{V}_{LZG} &= \sum_{g,h}\sum_{i,j} 1[G_i = G_j = g, H_i \neq H_j, i \neq j] E[\eta_i\eta_j | G_i = G_j = g, H_i \neq H_j, i \neq j] \\
&\quad + \sum_{g,h}\sum_{i,j} 1[G_i = G_j = g, H_i = H_j = h, i \neq j] E[\eta_i\eta_j | G_i = G_j = g, H_i = H_j = h, i \neq j] \\
&\quad + \sum_i E[\eta_i^2] \\
&= \sum_g (N_g^G)^2(\tilde{\nu}_g^2 + V_G(\tilde{\varepsilon}_g^w)^2) + \sum_{g,h}(N_{gh}^2)[(\tilde{\nu}_{gh}^2 + V_{GH}(\tilde{\varepsilon}_{gh}^w)^2) - (\tilde{\nu}_g^2 + V_G(\tilde{\varepsilon}_g^w)^2)] \\
&\quad + \sum_i [(1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2 - \nu_i^2 - V_{GH}(\varepsilon_i^w)^2]
\end{aligned}
$$

To obtain the CGM variance,

$$
\begin{aligned}
N^2\mu(1-\mu)\mathbb{V}_{CGM} &= N^2\mu(1-\mu)(\mathbb{V}_{LZG} + \mathbb{V}_{LZH} - \mathbb{A}\mathbb{V}_{LZGH}) \\
&= \sum_g (N_g^G)^2(\tilde{\nu}_g^2 + V_G(\tilde{\varepsilon}_g^w)^2) + \sum_{g,h}(N_{gh}^2)[(\tilde{\nu}_{gh}^2 + V_{GH}(\tilde{\varepsilon}_{gh}^w)^2) - (\tilde{\nu}_g^2 + V_G(\tilde{\varepsilon}_g^w)^2)] \\
&\quad + \sum_h (N_h^H)^2(\tilde{\nu}_h^2 + V_H(\tilde{\varepsilon}_h^w)^2) + \sum_{g,h}(N_{gh}^2)[(\tilde{\nu}_{gh}^2 + V_{GH}(\tilde{\varepsilon}_{gh}^w)^2) - (\tilde{\nu}_g^2 + V_G(\tilde{\varepsilon}_g^w)^2)] \\
&\quad - \sum_{g,h}(N_{gh}^2)(\tilde{\nu}_{gh}^2 + V_{GH}(\tilde{\varepsilon}_{gh}^w)^2) + \sum_i [(1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2 - \nu_i^2 - V_{GH}(\varepsilon_i^w)^2] \\
&= \sum_g (N_g^G)^2(\tilde{\nu}_g^2 + V_G(\tilde{\varepsilon}_g^w)^2) + \sum_h (N_h^H)^2(\tilde{\nu}_h^2 + V_H(\tilde{\varepsilon}_h^w)^2) \\
&\quad + \sum_{g,h}(N_{gh}^2)[(\tilde{\nu}_{gh}^2 + V_{GH}(\tilde{\varepsilon}_{gh}^w)^2) - (\tilde{\nu}_g^2 + V_G(\tilde{\varepsilon}_g^w)^2) - (\tilde{\nu}_h^2 + V_H(\tilde{\varepsilon}_h^w)^2)] \\
&\quad + \sum_i [(1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2 - \nu_i^2 - V_{GH}(\varepsilon_i^w)^2]
\end{aligned}
$$

□

Proof of Theorem 2. Use the result in Lemma 5 and Lemma 6 for the expression for the asymptotic variances. Then, there remains two parts to this proof. The first part deals with the convergence of the variance estimators, and the second part deals with $\mathbb{V}_\eta^{-1} Var(\hat{\tau}) = 1 + o_P(1)$.

Part I

It is sufficient to show the consistency of LZ estimators. If the LZ estimators are consistent, so is the CGM estimator by CMT. EHW is a special case of LZ. The rest of this derivation thus shows that the sample analog of the various conditional expectation objects used to derive Lemma 6 do indeed converge to those expectations.

Observe that convergence rate is driven by $\sum_g \frac{(N_g^G)^2}{N^2}$ (or on the h dimension, but I use g WLOG). Define r_N such that $r_N \sum_g \frac{(N_g^G)^2}{N^2} \to 1$. Everything else (e.g., $\sum_{g,h} N_{gh}/N$) has to converge faster than this object. For instance, if we have iid observations with everyone in their own cluster, $r_N = N$. Due to the definition of r_N and the fact that we have ruled out $\mathbb{V}_{LZG} = 0$, we can conclude there exists a constant $c > 0$ such that $r_N \mathbb{V}_{LZG} \to c$.

Since $\mathbb{V}_{LZG}^{-1} \hat{V}_{LZG} \xrightarrow{p} 1$ is equivalent to $\frac{r_N(\hat{V}_{LZG} - \mathbb{V}_{LZG})}{r_N \mathbb{V}_{LZG}} \xrightarrow{p} 0$, it is sufficient to show the latter. Further, since $r_N \mathbb{V}_{LZG} \to c > 0$, it is sufficient to show numerator is $o_P(1)$, i.e., $r_N(\hat{V}_{LZG} - \mathbb{V}_{LZG}) = o_P(1)$.

Observe that:

$$\begin{aligned}
&\frac{1}{N^2} \sum_{g,h} \sum_{i,j} 1[G_i = G_j = g, H_i = H_j = h] a_{G_i} a_{G_j} \\
&\quad = \frac{1}{N^2} \sum_{g,h} \sum_{i,j} 1[G_i = G_j = g, H_i = H_j = h] (Q_{G_i}^G - \mu) \tilde{\varepsilon}_i^w (Q_{G_j}^G - \mu) \tilde{\varepsilon}_j^w \\
&\quad = \frac{1}{N^2} \sum_{g,h} N_{gh}^2 (Q_g^G - \mu)^2 (\tilde{\varepsilon}_{gh}^w)^2
\end{aligned}$$

First observe that $\sum_g \left(\frac{N_g^G}{N}\right)^2 \to 0$ implies $\sum_g \left(\frac{N_g^G}{N}\right)^4 \to 0$ since $|\frac{N_g^G}{N}| \leq 1$. Subtract $\frac{1}{N^2} \sum_{g,h} N_{gh}^2 V_G (\tilde{\varepsilon}_{gh}^w)^2$ from the object above, and obtain:

$$\begin{aligned}
\sum_{g,h} \frac{N_{gh}^2}{N^2} [(Q_g^G - \mu)^2 - V_G] (\tilde{\varepsilon}_{gh}^w)^2 &= \sum_g \frac{(N_g^G)^2}{N^2} [(Q_g^G - \mu)^2 - V_G] \left(\sum_h \frac{N_{gh}^2}{(N_g^G)^2} (\tilde{\varepsilon}_{gh}^w)^2 \right) \\
&= \sum_g \frac{(N_g^G)^2}{N^2} [(Q_g^G - \mu)^2 - V_G] O(1)
\end{aligned}$$

Scaling this by r_N,

$$r_N \sum_{g,h} \frac{N_{gh}^2}{N^2} [(Q_g^G - \mu)^2 - V_G] (\tilde{\varepsilon}_{gh}^w)^2 = \sum_g \frac{r_N (N_g^G)^2}{N^2} [(Q_g^G - \mu)^2 - V_G] O(1) = o_P(1) \tag{8}$$

Here, weights are given by $\frac{r_N (N_g^G)^2}{N^2}$. Since $r_N \sum_g \frac{(N_g^G)^2}{N^2} \to 1$, by using Holder inequality,

$$\begin{aligned}
\sum_g \frac{r_N^2 (N_g^G)^4}{N^4} &= \sum_g \frac{N_g^G}{N} \frac{r_N^2 (N_g^G)^3}{N^3} \\
&\leq \left(\sum_g \frac{(N_g^G)^2}{N^2} r_N \right) \left(\sum_g \frac{(N_g^G)^2}{N^2} r_N \right) \left(\sum_g \frac{(N_g^G)^2}{N^2} \right) \\
&= (1 + o(1))^2 o(1) = o(1)
\end{aligned}$$

Then, LLN applies, yielding the $o_P(1)$ term in Equation 8. The derivation for all other terms is fairly analogous.

Next, we know $E[(Q_{G_i}^G - \mu)(Q_{G_j}^G - \mu) | G_i \neq G_j] = 0$.

$$\begin{aligned}
&\frac{r_N}{N^2} \sum_h \sum_{i,j} 1[G_i \neq G_j, H_i = H_j = h](Q_{G_i}^G - \mu)(Q_{G_j}^G - \mu) \\
&= \frac{r_N}{N^2} \sum_{g,h} \left[\sum_{i,j} 1[G_i = g, H_i = h](Q_{G_i}^G - \mu)\varepsilon_i^w 1[G_j \neq g, H_j = h](Q_{G_j}^G - \mu)\varepsilon_j^w \right] \\
&= \frac{r_N}{N^2} \sum_h (N_h^H)^2 \left[\sum_g \frac{N_{gh}}{N_h^H} (Q_g^G - \mu)\tilde{\varepsilon}_g^w \sum_{g' \neq g} \frac{N_{gh}}{N_h^H} (Q_{g'}^G - \mu)\tilde{\varepsilon}_{g'}^w \right] \\
&= \frac{r_N}{N^2} \sum_h (N_h^H)^2 \left[\sum_g \frac{N_{gh}}{N_h^H} (Q_g^G - \mu)\tilde{\varepsilon}_g^w o_P(1) \right] \\
&= \frac{r_N}{N^2} \sum_h (N_h^H)^2 o_P(1) \\
&= o_P(1)
\end{aligned}$$

Convergence occurs by Assumption 3 which yields $\sum_{g' \neq g} (\frac{N_{gh}}{N_h^H})^2 \to 0$, so that LLN can be applied. The subsequent steps are analogous.

These results imply:

$$\frac{1}{N^2}\sum_{g,h}\sum_{i,j}1[G_i=G_j=g,H_i\neq H_j=h]b_{H_i}b_{H_j}=o_P(1)$$
$$\frac{1}{N^2}\sum_{g,h}\sum_{i,j}1[G_i=G_j=g,H_i\neq H_j=h]b_{H_i}c_{G_jH_j}=o_P(1)$$

Consider $a_{G_i}c_{G_jH_j}$:

$$\begin{aligned}
&\frac{1}{N^2}\sum_{g,h}\sum_{i,j}1[G_i=G_j=g,H_i=H_j=h]a_{G_i}c_{G_jH_j}\\
&\quad=\frac{1}{N^2}\sum_{g,h}\sum_{i,j}1[G_i=G_j=g,H_i=H_j=h](Q^G_{G_i}-\mu)\varepsilon^w_i(Q^{GH}_{G_jH_j}-Q^G_{G_j}-Q^H_{H_j}+\mu)\varepsilon^w_j\\
&\quad=\frac{1}{N^2}\sum_{g,h}N^2_{gh}(Q^G_g-\mu)(Q^{GH}_{gh}-Q^G_g)(\tilde{\varepsilon}^w_{gh})^2-\frac{1}{N^2}\sum_{g,h}N^2_{gh}(Q^G_g-\mu)(Q^H_h-\mu)(\tilde{\varepsilon}^w_{gh})^2
\end{aligned}$$

First observe that, using the same arguments as before,

$$\begin{aligned}
\frac{r_N}{N^2}\sum_{g,h}N^2_{gh}(Q^G_g-\mu)(Q^H_h-\mu)(\tilde{\varepsilon}^w_{gh})^2&=\frac{r_N}{N^2}\sum_g(Q^G_g-\mu)\sum_h N^2_{gh}(Q^H_h-\mu)\\
&=\frac{r_N}{N^2}\sum_g(N^G_g)^2(Q^G_g-\mu)\sum_h\frac{N^2_{gh}}{(N^G_g)^2}(Q^H_h-\mu)=o_P(1)
\end{aligned}$$

Hence, by adding and subtracting μ in $(Q^{GH}_{gh}-Q^G_g)$,

$$\begin{aligned}
&\frac{r_N}{N^2}\sum_{g,h}\sum_{i,j}(1[G_i=G_j=g,H_i=H_j=h]a_{G_i}c_{G_jH_j})-\frac{r_N}{N^2}\sum_{g,h}N^2_{gh}(V_{CG}-V_G)(\tilde{\varepsilon}^w_{gh})^2\\
&\quad=\frac{r_N}{N^2}\sum_{g,h}N^2_{gh}[(Q^G_g-\mu)(Q^{GH}_{gh}-\mu)-V_{CG}](\tilde{\varepsilon}^w_{gh})^2-\frac{r_N}{N^2}\sum_{g,h}N^2_{gh}[(Q^G_g-\mu)^2-V_G](\tilde{\varepsilon}^w_{gh})^2
\end{aligned}$$

We know $\frac{r_N}{N^2}\sum_{g,h}N^2_{gh}[(Q^G_g-\mu)^2-V_G](\tilde{\varepsilon}^w_{gh})^2=o_P(1)$ due to the derivation from before. The first

term converges since $(Q_g^G - \mu)(Q_{gh}^{GH} - \mu)$ are independent over g conditional on h.

$$\frac{r_N}{N^2}\sum_{g,h} N_{gh}^2[(Q_g^G - \mu)(Q_{gh}^{GH} - \mu) - V_{CG}](\tilde{\varepsilon}_{gh}^w)^2$$
$$= \frac{r_N}{N^2}\sum_h (N_h^H)^2 \sum_g \frac{N_{gh}^2}{(N_h^H)^2}[(Q_g^G - \mu)(Q_{gh}^{GH} - \mu) - V_{CG}](\tilde{\varepsilon}_{gh}^w)^2 = o_P(1)$$

Convergence occurs due to Assumption 3 using the same arguments as before, giving weights that are well-behaved and finite moments. Hence,

$$\frac{r_N}{N^2}\sum_{g,h}\sum_{i,j} 1[G_i = G_j = g, H_i = H_j = h]a_{G_i}c_{G_jH_j} = \frac{r_N}{N^2}\sum_{g,h} N_{gh}^2(V_{CG} - V_G)(\tilde{\varepsilon}_{gh}^w)^2 + o_P(1)$$

Analogously, we can obtain:

$$\frac{r_N}{N^2}\sum_{g,h}\sum_{i,j} 1[G_i = G_j = g, H_i = H_j = h]c_{G_iH_i}c_{G_jH_j}$$
$$= \frac{r_N}{N^2}\sum_{g,h} N_{gh}^2(V_{GH} + V_G + V_H - 2V_{CG} - 2V_{CH})(\tilde{\varepsilon}_{gh}^w)^2 + o_P(1)$$

Combine these results using CMT to get convergence of the LZ estimator.

Part II

Consider the OLS Estimator:

$$\hat{\tau} - \tau = \frac{\frac{1}{N}\sum_i (W_i - \bar{W})(\varepsilon_i - \bar{\varepsilon})}{\frac{1}{N}\sum_i (W_i - \bar{W})^2} = \frac{\frac{1}{N}\sum_i (W_i - \mu)(\varepsilon_i - \bar{\varepsilon})}{\frac{1}{N}\sum_i (W_i - \bar{W})^2}$$
$$= \sqrt{\mu(1-\mu)}\frac{\frac{1}{N}\sum_i \eta_i + \frac{1}{N\sqrt{\mu(1-\mu)}}\sum_i (W_i - \mu)\bar{\varepsilon}}{\frac{1}{N}\sum_i (W_i - \bar{W})^2}$$

The denominator converges to $\mu(1-\mu)$. In the numerator, $\bar{\varepsilon}$ and $\frac{1}{N}\sum_i (W_i - \mu)$ both converge at rate r_N, so the dominating term is $\frac{1}{N}\sum_i \eta_i$. Then, by CMT, $\mathbb{V}_\eta^{-1} Var(\hat{\tau}) = 1 + o_P(1)$.

□

Proof of Corollary 1. The difference between the CGM variance and the true variance (multiplied

by N) is:

$$N^2\mu(1-\mu)(\mathbb{V}_{CGM}-\mathbb{V}_\eta) = \sum_g (N_g^G)^2(\tilde{\nu}_g^2) + \sum_h (N_h^H)^2(\tilde{\nu}_h^2) + \sum_{g,h}(N_{gh}^2)[(\tilde{\nu}_{gh}^2) - \tilde{\nu}_g^2 - \tilde{\nu}_h^2]$$
$$= \sum_g \left((N_g^G)^2 - \sum_h N_{gh}^2\right)(\tilde{\nu}_g^2) + \sum_h \left((N_h^H)^2 - \sum_g N_{gh}^2\right)(\tilde{\nu}_h^2) + \sum_{g,h}(N_{gh}^2)(\tilde{\nu}_{gh}^2) \geq 0$$

We can calculate similar objects for LZ and EHW variances:

$$N^2\mu(1-\mu)(\mathbb{V}_{LZG}-\mathbb{V}_\eta) = \sum_g (N_g^G)^2(\tilde{\nu}_g^2) - \sum_h (N_h^H)^2 V_H(\tilde{\varepsilon}_h^w)^2$$
$$+ \sum_{g,h}(N_{gh}^2)[(\tilde{\nu}_{gh}^2 + V_{GH}(\tilde{\varepsilon}_{gh}^w)^2) - (\tilde{\nu}_g^2 + V_G(\tilde{\varepsilon}_g^w)^2)] + \sum_i [(1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2 - \nu_i^2 - V_{GH}(\varepsilon_i^w)^2]$$
$$- \sum_{g,h}(N_{gh}^2)[V_{GH}(\tilde{\varepsilon}_{gh}^w)^2 - V_G(\tilde{\varepsilon}_g^w)^2 - V_H(\tilde{\varepsilon}_h^w)^2] - \sum_i [(1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2 - \nu_i^2 - V_{GH}(\varepsilon_i^w)^2]$$
$$= \sum_g (N_g^G)^2(\tilde{\nu}_g^2) - \sum_h (N_h^H)^2 V_H(\tilde{\varepsilon}_h^w)^2 + \sum_{g,h}(N_{gh}^2)[(\tilde{\nu}_{gh}^2 + V_{GH}(\tilde{\varepsilon}_{gh}^w)^2) - (\tilde{\nu}_g^2 + V_G(\tilde{\varepsilon}_g^w)^2)]$$
$$- \sum_{g,h}(N_{gh}^2)[V_{GH}(\tilde{\varepsilon}_{gh}^w)^2 - V_G(\tilde{\varepsilon}_g^w)^2 - V_H(\tilde{\varepsilon}_h^w)^2]$$
$$= \sum_g (N_g^G)^2(\tilde{\nu}_g^2) - \sum_h (N_h^H)^2 V_H(\tilde{\varepsilon}_h^w)^2 + \sum_{g,h}(N_{gh}^2)[\tilde{\nu}_{gh}^2 - \tilde{\nu}_g^2 + V_H(\tilde{\varepsilon}_h^w)^2]$$

$$N^2\mu(1-\mu)(\mathbb{V}_{EHW}-\mathbb{V}_\eta) = \sum_i (1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2 - \sum_g (N_g^G)^2 V_G(\tilde{\varepsilon}_g^w)^2 - \sum_h (N_h^H)^2 V_H(\tilde{\varepsilon}_h^w)^2$$
$$- \sum_{g,h}(N_{gh}^2)[V_{GH}(\tilde{\varepsilon}_{gh}^w)^2 - V_G(\tilde{\varepsilon}_g^w)^2 - V_H(\tilde{\varepsilon}_h^w)^2] - \sum_i [(1-\mu)\varepsilon_i(1)^2 + \mu\varepsilon_i(0)^2 - \nu_i^2 - V_{GH}(\varepsilon_i^w)^2]$$
$$= -\sum_g (N_g^G)^2 V_G(\tilde{\varepsilon}_g^w)^2 - \sum_h (N_h^H)^2 V_H(\tilde{\varepsilon}_h^w)^2 - \sum_{g,h}(N_{gh}^2)[V_{GH}(\tilde{\varepsilon}_{gh}^w)^2 - V_G(\tilde{\varepsilon}_g^w)^2 - V_H(\tilde{\varepsilon}_h^w)^2]$$
$$+ \sum_i [\nu_i^2 + V_{GH}(\varepsilon_i^w)^2]$$

□

A.3 Proofs for Section 4.3

Proof of Lemma 4. First, it can be shown that Assumption 5(i) implies that, as $N \to 0$, there exists $c > 0$ such that $\sum_g \left(\tilde{\varepsilon}_g^w\right)^2 \left(\frac{N_g^G}{N}\right)^2 \geq c \sum_g \left(\frac{N_g^G}{N}\right)^2$ or $\sum_h \left(\tilde{\varepsilon}_h^w\right)^2 \left(\frac{N_h^H}{N}\right)^2 \geq c \sum_h \left(\frac{N_h^H}{N}\right)^2$. It is sufficient to make the argument on the g dimension.

For notational convenience, let $Y_g(w) = \frac{1}{N_g^G} \sum_{i \in S_g} Y_i(w)$. Since there are finitely many g such that $Y_g(w) = \bar{Y}(w)$ for all w, there are infinitely many g such that $(Y_g(w) - \bar{Y}(w))^2 > 0$ for at least one w by Assumption 2(i). Since we have defined $\tilde{\varepsilon}_g^w := \left(\sqrt{\frac{1-\mu}{\mu}}\tilde{\varepsilon}_g(1) + \sqrt{\frac{\mu}{1-\mu}}\tilde{\varepsilon}_g(0)\right)$, $(Y_g(w) - \bar{Y}(w))^2 > 0$ for some g, w implies that $(\tilde{\varepsilon}_g^w)^2 > 0$ for that given g. Consequently, there are finitely many g such that $(\tilde{\varepsilon}_g^w)^2 = 0$ and infinitely many g such that $(\tilde{\varepsilon}_g^w)^2 > 0$. Then, select $c = \min_{g:(\tilde{\varepsilon}_g^w)^2 > 0} (\tilde{\varepsilon}_g^w)^2$. By Assumption 2(ii), there exists $c > 0$ such that $\sum_g \left(\tilde{\varepsilon}_g^w\right)^2 \left(\frac{N_g^G}{N}\right)^2 \geq c \sum_g \left(\frac{N_g^G}{N}\right)^2$.

Set $\delta = 2$. For arbitrary constants C, c,

$$\begin{aligned}
\lim_{N\to\infty} \frac{\sum_g E\left[\left(\frac{(N_g^G)}{N}\tilde{\varepsilon}_g^w (Q_g^G - \mu)\right)^4\right]}{\left(\sum_g (\tilde{\varepsilon}_g^w)^2 \left(\frac{N_g^G}{N}\right)^2\right)^2} &= \lim_{N\to\infty} \frac{\sum_g \frac{(N_g^G)^4}{N^4} C}{\left(\sum_g (\tilde{\varepsilon}_g^w)^2 \left(\frac{N_g^G}{N}\right)^2\right)^2} \\
&\leq \lim_{N\to\infty} \frac{\sum_g \frac{(N_g^G)^4}{N^4} C}{c\left(\sum_g \left(\frac{N_g^G}{N}\right)^2\right)^2} \\
&= C' \lim_{N\to\infty} \sum_g \left(\frac{\left(\frac{N_g^G}{N}\right)^2}{\sum_g \left(\frac{N_g^G}{N}\right)^2}\right)^2 = C' \lim_{N\to\infty} \sum_g \left(\frac{(N_g^G)^2}{\sum_g (N_g^G)^2}\right)^2 = 0
\end{aligned}$$

The last equality follows from Assumption 5(ii). □

Proof of Theorem 3. Consider the OLS Estimator:

$$\hat{\tau} - \tau = \sqrt{\mu(1-\mu)} \frac{\frac{1}{N}\sum_i \eta_i + \frac{1}{N\sqrt{\mu(1-\mu)}} \sum_i (W_i - \mu)\bar{\varepsilon}}{\frac{1}{N}\sum_i (W_i - \bar{W})^2}$$

Apply Slutsky argument, so we can focus on the numerator. It turns out that there exists R_N such that $R_N \bar{\varepsilon} = o_P(1)$ and $R_N \frac{1}{N}\sum_i (W_i - \mu) = o_P(1)$ and $s_N^{-1} o_P(R_N^{-2}) = o_P(1)$.[5] Then, numerator

[5] Consider iid data X_i with mean zero. By standard arguments, we know $\bar{X} := \frac{1}{N}\sum_i X_i = o_P(1)$ and $\sqrt{N}\bar{X} = O_P(1)$. Then, for $R_N \in (1, N^{1/2})$, we can get $R_N \bar{X} = o_P(1)$. This means $\bar{X} = o_P(R_N^{-1})$. While $\sqrt{N} o_P(R_N^{-1}) \neq o_P(1)$, it's

becomes $\frac{1}{N}\sum_i \eta_i + o_P(R_N^{-2})$, and it is sufficient to establish that $\frac{1}{s_N N}\sum_i \eta_i$ is asymptotically normal.

Using the same argument as Theorem 1, $\frac{1}{s_N}\sum_{i=1}^N \eta_i = \frac{1}{s_N}\sum_{i=1}^N \tilde{\eta}_i$, and $\tilde{\eta}_i$ each has mean zero. The Aldous-Hoover decomposition is hence identical to that of Theorem 1. Namely,

$$
\begin{aligned}
\tilde{\eta}_i &= a_{G_i} + b_{H_i} + c_{G_i H_i} + e_i \\
a_g &:= (Q_g^G - \mu)\tilde{\varepsilon}_g^w \\
b_h &:= (Q_h^H - \mu)\tilde{\varepsilon}_h^w \\
c_{gh} &:= (Q_{gh}^{GH} - Q_g^G - Q_h^H + \mu)\tilde{\varepsilon}_{gh}^w \\
e_i &:= \tilde{\eta}_i - Q_{G_i H_i}^{GH}\tilde{\varepsilon}_i^w
\end{aligned}
$$

Recall the sum of standard errors as the following:

$$
\begin{aligned}
s_N &:= \max\{s_N^G, s_N^H\} \\
(s_N^G)^2 &:= \sum_g V_G\left(\tilde{\varepsilon}_g^w\right)^2 \left(\frac{N_g^G}{N}\right)^2 \\
(s_N^H)^2 &:= \sum_h V_H\left(\tilde{\varepsilon}_h^w\right)^2 \left(\frac{N_h^H}{N}\right)^2
\end{aligned}
$$

Consider the case where s_N is nonzero. Apply the Lindeberg Central Limit Theorem, which states:

(Lindeberg CLT). Suppose $\{X_1, \cdots, X_n\}$ is a sequence of independent random variables, each with finite expected value μ_i and variance σ_i^2. Define $s_n^2 = \sum_{i=1}^n \sigma_i^2$. If for every $r > 0$, $\lim_{n\to\infty} \frac{1}{s_n^2}\sum_{i=1}^n E[(X_i - \mu_i)^2 1\{|X_i - \mu_i| > r s_n\}] = 0$. Then, $\frac{1}{s_n}\sum_{i=1}^n (X_i - \mu_i) \xrightarrow{d} N(0,1)$.

The Lyapunov condition states that for some $\delta > 0$, $\lim_{n\to\infty} \frac{1}{s_n^{2+\delta}}\sum_{i=1}^n E[|X_i - \mu_i|^{2+\delta}] = 0$. Since the Lyapunov condition implies the Lindeberg condition, it is sufficient to verify the Lyapunov condition for $\delta > 0$. Recall that $E[(Q_g^G - \mu)^{2+\delta}]$ is finite due to binary assignment. Then, using Assumption 4, and C to denote an arbitrary constant,

possible to select R_N^{-1} (e.g., $R_N = N^{1/3}$) so that $\sqrt{N} o_P(R_N^{-2}) \neq o_P(1)$. Extend this reasoning to our nonstandard convergence.

$$\lim_{N\to\infty} \frac{\sum_g E\left[\left(\frac{(N_g^G)}{N}\tilde{\varepsilon}_g^w(Q_g^G-\mu)\right)^{2+\delta}\right]}{\left(\sum_g V_G(\tilde{\varepsilon}_g^w)^2\left(\frac{N_g^G}{N}\right)^2\right)^{(2+\delta)/2}} = \lim_{N\to\infty} \frac{\sum_g \left(\frac{N_g^G}{N}\right)^{2+\delta} C}{\left(\sum_g (\tilde{\varepsilon}_g^w)^2\left(\frac{N_g^G}{N}\right)^2\right)^{(2+\delta)/2}} = 0$$

Then, by Lindeberg CLT,

$$\frac{1}{s_N^G}\sum_g a_g \frac{N_g^G}{N} \xrightarrow{d} N(0,1)$$
$$\frac{1}{s_N^H}\sum_h b_h \frac{N_h^H}{N} \xrightarrow{d} N(0,1)$$

The two convergent sums are independent of each other because their only source of randomness is on Q^G and Q^H, which are independent. Since the problem is symmetric regardless of whether s_N^G or s_N^H is larger, we can consider the case where $s_N = s_N^G \geq s_N^H$ WLOG. Let $s_N^H/s_N^G \to r$, with $|r| < 1$, and r is allowed to be zero. Then, using Z to denote $N(0,1)$,

$$\frac{1}{s_N^G}\sum_h b_h \frac{N_h^H}{N} = \frac{s_N^H}{s_N^G}\frac{1}{s_N^H}\sum_h b_h \frac{N_h^H}{N} \xrightarrow{d} rZ$$

If $r = 0$, then the above object converges in probability to zero. Since $|r| \leq 1$, we either have a sum of two independent asymptotically normal objects, so their sum will also be asymptotically normal. Hence, $\frac{1}{s_N^G}\left(\sum_g N_g a_g + \sum_h N_h b_h\right) \xrightarrow{d} Z$. Finally, it remains to show that the remaining terms c_{gh} and e_i are indeed $o_P(1)$.

Since potential outcomes are bounded and assignment is binary, c_{gh} is L^2 integrable. Then, following the argument in Menzel (2021), the spectral representation theorem implies that there exists a representation:

$$c_{gh} = \sum_{k=1}^{\infty} c_k \phi_k(A_g)\psi_k(B_h)$$

where $\{c_k\}$ is a sequence of constants that is square summable. Then,

$$\begin{aligned}
\frac{1}{s_N^G}\sum_{g,h}\frac{N_{gh}}{N}c_{gh} &= \frac{1}{s_N^G}\sum_{g,h}\frac{N_{gh}}{N}\sum_{k=1}^{\infty}c_k\phi_k(A_g)\psi_k(B_h) \\
&= \sum_k c_k\left(\sum_g \frac{(\frac{N_g^G}{N})}{s_N^G}\phi_k(A_g)\right)\left(\sum_h \frac{N_{gh}}{N_g^G}\psi_k(B_h)\right) \\
&= \sum_k c_k O_P(1)o_P(1) = o_P(1)
\end{aligned}$$

In the last step, $\sum_g \frac{(\frac{N_g^G}{N})}{s_N^G}\phi_k(A_g) = O_P(1)$ is given by the Lindeberg CLT, and the Lindeberg condition can be verified as before. $\sum_h \frac{N_{gh}}{N_g^G}\psi_k(B_h) = o_P(1)$ by LLN. The infinite sum over k yields $o_P(1)$ due to the square summability condition. By a similar argument,

$$\frac{1}{s_N^G}\sum_i e_i = \sum_k \tilde{c}_k O_P(1)o_P(1) = o_P(1)$$

Now, we have $\frac{1}{s_N}\sum_{i=1}^N \frac{\eta_i}{N} \xrightarrow{d} N(0,\Omega)$. Recall from Equation 4 that $\hat{\tau}-\tau = \frac{1}{N\sqrt{\mu(1-\mu)}}\sum_{i=1}^N \eta_i + o_P(1)$. Then, for $Z \sim N(0,\Omega)$,

$$\begin{aligned}
\frac{1}{s_N}\frac{1}{\sqrt{\mu(1-\mu)}}\sum_i \frac{\eta_i}{N} &\overset{a}{\sim} \frac{Z}{\sqrt{\mu(1-\mu)}} \\
\frac{1}{s_N}(\hat{\tau}-\tau) &\overset{a}{\sim} \frac{Z}{\sqrt{\mu(1-\mu)}} \\
\frac{\sqrt{\mu(1-\mu)}}{s_N}(\hat{\tau}-\tau) &\overset{a}{\sim} N(0,\Omega)
\end{aligned}$$

□

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